

Heat of Hydration of Portland and Composite Cements Under Semi-Adiabatic Conditions

by

Abdur Nur Tushar & Zisan Zia

ID: 2004061

ID: 2004153



Department of Civil Engineering, Bangladesh University of Engineering and
Technology
Dhaka, Bangladesh

June, 2026

Heat of Hydration of Portland and Composite Cements Under Semi-Adiabatic Conditions

by

Abdur Nur Tushar & Zisan Zia

ID: 2004061

ID: 2004153

A thesis submitted to the Department of Civil Engineering
in partial fulfillment of the degree of BACHELOR OF SCIENCE IN CIVIL
ENGINEERING



Department of Civil Engineering, Bangladesh University of Engineering and
Technology
Dhaka, Bangladesh

June 2026

BOARD OF EXAMINERS

The thesis titled “Heat of Hydration of Portland and Composite Cements Under Semi-Adiabatic Conditions” submitted by Abdur Nur Tushar, Student No.:2004061 & Zisan Zia, Student No.: 2004153, has been accepted as satisfactory in partial fulfillment of the requirement for the degree of Bachelor of Science in Civil Engineering on 13 June 2026.

Dr. Raquib Ahsan

Professor

Department of Civil Engineering, BUET,
Dhaka-1000

Chairman(Super-visor)

Dr. Nazrul Islam

Professor

Department of Civil Engineering, BUET,
Dhaka-1000

Member

Dr. S. M. Faisal Mahmood, Associate

Associate Professor

Department of Civil Engineering, BUET,
Dhaka-1000

Member

DECLARATION

We hereby declare that this thesis is our own work and, to the best of our knowledge, it contains no materials previously published or written by another person, or has been accepted for the award of any other degree or diploma at Bangladesh University of Engineering and Technology (BUET) or any other educational institution, except where due acknowledgement is made. We also declare that the intellectual content of this thesis is the product of our own work and any contribution made to the research by others, with whom We have worked at BUET or elsewhere, is explicitly acknowledged.

Abdur Nur Tushar

Zisan Zia

Dedicated

to

“Our Parents”

*For our parents,
who taught us to
chase the stars, even
when the night
seemed endless.*

ACKNOWLEDGEMENTS

We extend our deepest gratitude to the Almighty Allah for His boundless kindness and grace, granting us the strength and perseverance to complete this study.

We are profoundly grateful to our family for their unwavering love and support. Their constant encouragement has been instrumental in all our achievements. Our heartfelt appreciation goes to our parents, whose guidance and unwavering belief in us have shaped our journey.

We wish to express our sincere gratitude to our esteemed supervisor, Dr. Raquib Ahsan (Professor, Department of Civil Engineering, BUET). His invaluable guidance, insightful feedback, and steadfast support throughout this research were crucial to its completion.

We are also deeply thankful to all our teachers for their dedication and efforts in imparting knowledge.

ABSTRACT

Heat generated during cement hydration is a critical factor influencing temperature rise and the risk of thermal cracking in mass concrete structures. Reliable prediction of early-age temperature development requires accurate characterization of hydration kinetics for commonly used cement types under controlled conditions. This study investigates the hydration behavior of four commercially available cements conforming to BDS EN 197-1:2003: CEM I 52.5N, CEM II A-M 42.5N, CEM II A-M 42.5R, and CEM II B-M 42.5N, tested under semi-adiabatic and sealed conditions. The inclusion of the rapid-hardening variant CEM II A-M 42.5R alongside its standard-strength counterpart enables a direct assessment of how early strength gain characteristics influence overall hydration heat evolution.

Mineralogical characterization was performed using X-ray diffraction (XRD) to identify major clinker phases, including C_3S , C_2S , C_3A , and C_4AF . Hydration heat evolution was monitored by semi-adiabatic calorimetry, from which the heat generation rate and cumulative heat release were determined for each cement type. A MATLAB-based two-component Gaussian model was applied to represent the multi-stage hydration process by separately capturing the early aluminate reaction and the dominant silicate hydration peak. In addition, cumulative heat evolution was modeled using sixth-order polynomial functions to produce continuous analytical expressions suitable for numerical thermal simulations. Experimental observations were further validated using hydration predictions from the NIST Virtual Cement and Concrete Testing Laboratory (VCCTL).

Results show clear differences in thermal behavior among the four tested cements. CEM I 52.5N exhibited the highest cumulative heat release (~ 467 J/g) and the fastest overall hydration kinetics, indicating the greatest thermal cracking potential in mass concrete applications. CEM II A-M 42.5R demonstrated notably accelerated early-age heat evolution relative to the standard CEM II A-M 42.5N, reflecting its higher reactive phase content and finer grinding, though its long-term cumulative heat remained moderate. CEM II A-M 42.5N showed intermediate thermal behavior between the high-heat and low-heat cement. In contrast, CEM II B-M 42.5N exhibited the lowest cumulative heat release (~ 255 J/g) and the slowest heat evolution, suggesting the most favorable thermal performance for temperature-sensitive concrete placements.

The developed hydration models provide practical input parameters for thermal analysis, cement selection, and temperature-control strategies in mass concrete structures, with particular emphasis on distinguishing the role of rapid-hardening cements in early-age thermal risk assessment.

TABLE OF CONTENTS

CONTENTS

<i>Declaration</i>	4
<i>Acknowledgements</i>	6
<i>Abstract</i>	7
<i>TABLE OF CONTENTS</i>	8
<i>LIST OF FIGURES</i>	12
<i>LIST OF TABLES</i>	14
<i>Notation</i>	15
CHAPTER 1	17
<i>Introduction</i>	17
1.1 Background	17
1.2 Significance of the Study	19
1.3 Objectives of the Research	20
1.4 Scope of the Study	20
1.5 Organization of Thesis	22
CHAPTER 2	23
<i>Literature review</i>	23
2.1 Introduction	23
2.2 Cement Hydration Mechanisms	23
2.3 Heat Evolution During Cement Hydration	24
2.4 Calorimetric Techniques for Hydration Analysis	24
2.4.1 Isothermal Calorimetry	25
2.4.2 Semi-Adiabatic Calorimetry	25
2.4.3 Hydration Behavior of Portland and Composite Cements	25
2.5 Hydration Kinetics and Analytical Modeling	26
2.6 X-Ray Diffraction Characterization of Cement	27
2.7 Numerical Simulation Using VCCTL	27
2.8 Research Gaps	27
CHAPTER 3	29
<i>Methodology</i>	29

3.1 Introduction	29
3.2 Methodology Overview	30
3.3 Materials Used	32
3.4 Sample Identification	34
3.5 Cement Paste Preparation	35
3.6 Mineralogical Characterization by X-Ray Diffraction	36
3.6.1 Purpose of XRD Analysis	36
3.6.2 XRD Instrumentation and Test Conditions	37
3.6.3 Phase Identification and Semi-Quantitative Estimation	38
3.7 Semi-Adiabatic Calorimetry	39
3.7.1 Purpose of Semi-Adiabatic Calorimetry	39
3.7.2 Semi-Adiabatic Calorimeter Setup	40
3.7.3 Calibration of the Calorimeter	41
3.7.4 Hydration Test Procedure	43
3.8 Heat Flow Calculation	44
3.8.1 Governing Heat Balance Equation	44
3.8.2 Heat Generation Rate per Unit Mass	46
3.8.3 Cumulative Heat Release	46
3.9 Degree of Hydration and Variable Heat Capacity	47
Degree of hydration [r]	49
3.10 MATLAB-Based Data Processing	50
3.10.2 Output of Data Processing	51
3.10.3 Two -Component Gaussian Model	51
3.10.4 Polynomial Fitting of Cumulative Heat	52
3.11 VCCTL Simulation	53
3.11.1 Purpose of VCCTL Simulation	53
3.11.2 VCCTL Input Parameters	53
3.11.3 VCCTL Output and Comparison	54
3.12 Validation Criteria	54
3.13 Quality Control Measures	55
3.14 Limitations of Methodology	56
3.15 Summary	56
CHAPTER 4	58
Results and Discussions	58

4.1 Introduction	58
4.2 XRD Analysis and Mineralogical Characterization	58
4.3 Semi-Adiabatic Calorimetric Measurements	62
4.4 VCCTL Simulation and Validation	65
4.5 Discussion	67
4.5.1 Clinker Composition — Hydration Kinetics Correlation	67
4.5.2 Thermal Cracking Risk Assessment	68
4.5.3 Polynomial Model Applicability	70
4.5.4 Limitations and Future Work	70
4.6 Chapter Summary	70
CHAPTER 5	71
Conclusions and Future work	<i>Error! Bookmark not defined.</i>
5.1 General	Error! Bookmark not defined.
5.2 Limitations of the Study	Error! Bookmark not defined.
5.3 Conclusions	Error! Bookmark not defined.
5.4 Practical Implications	Error! Bookmark not defined.
5.5 Recommendations and Future Work	Error! Bookmark not defined.
5.6 Proposed Framework for Cement Selection in Mass Concrete	Error! Bookmark not defined.
5.7 Closing Remarks	Error! Bookmark not defined.
REFERENCES	77
APPENDIX	80
Appendix A	81
A.1 General	81
A.2 XRD Measurement Conditions	81
A.3 XRD Diffractograms	82
A.4 XRD Phase Identification Results	84
Appendix B	88
B.1 General	88
B.2 Effective Heat Capacity of the Calorimeter System	88
B.3 Heat-Loss Coefficient from Cooling Experiment	89
Appendix C	92
C.1 Heat Balance Equation	92

C.2 Heat Generation Rate Per Unit Mass	92
C.3 Cumulative Heat Release	93
C.4 Degree of Hydration	93
Appendix D	95
D.1 General	95
D.2 CEM II B-M 42.5 N	95
D.3 CEM II A-M 42.5 N	96
D.4 CEM II A-M 42.5 R	97
D.5 CEM I 52.5 N	97
D.6 Summary of Gaussian Model Parameters	98
Appendix E	99
E.1 General	99
E.2 CEM II B-M 42.5 N	99
E.3 CEM II A-M 42.5 N	100
E.4 CEM II A-M 42.5 R	100
E.5 CEM I 52.5 N	101
E.6 VCCTL Simulation Summary	101
Appendix K	102
K.1 General	102
K.2 Algorithm Description	102
K.3 Step-by-Step Algorithm	103
K.4 MATLAB Code Used for Hydration Heat Analysis	106

LIST OF FIGURES

Figure 3.1: Overall Methodology Flowchart	30
Figure 3.2: Detailed Research Workflow Flowchart	32
Figure 3.3: Cement samples used in the experimental program	34
Figure 3.4: Preparation of cement paste specimen for semi-adiabatic Calorimetry	36
Figure 3.5: X-ray diffraction equipment used for mineralogical characterization.	37
Figure 3.6: Preparation of powdered cement sample for XRD analysis	38
Figure 3.7: XRD analysis workflow chart	39
Figure 3.8: Semi-adiabatic calorimeter used for hydration heat measurement	40
Figure 3.9: Schematic diagram of calorimeter setup	41
Figure 3.10: Calorimeter calibration flowchart	43
Figure 3.11: Temperature monitoring and data acquisition during semi-adiabatic calorimetry	44
Figure 3.12: Heat calculation flowchart	47
Figure 3.13: Specific heat of hardening cement paste	47
Figure 3.14: MATLAB data processing workflow	51
Figure 3.15: Gaussian fitting methodology chart	52
Figure 3.16: Polynomial fitting methodology chart	53
Figure 3.17: VCCTL simulation workflow chart	54
Figure 4.1: XRD diffraction pattern of Cement 1 (CEM II B-M 42.5 N)	59
Figure 4.2: XRD diffraction pattern of Cement 2 (CEM II A-M 42.5 N)	59
Figure 4.3: XRD diffraction pattern of Cement 3 (CEM II A-M 42.5 R)	60
Figure 4.4: XRD diffraction pattern of Cement 4 (CEM I 52.5 N)	60
Figure 4.5: Bar chart showing semi-quantitative XRD phase composition	61
Figure 4.6: MATLAB heat generation rate for all cement samples (Experimental)	63
Figure 4.7: MATLAB heat generation rate for all cement samples (Fitted Data – Gauss 2)	63
Figure 4.8 Combined MATLAB cumulative heat generation for all cement samples experimental (left) and two-component Gaussian (Gauss-2) fit (right).	64

Figure 4.9: Combined VCCTL-simulated degree of hydration versus time for all four cements (isothermal, $w/c = 0.40$). 66

Figure 4.10: Combined VCCTL-simulated cumulative heat release versus time for all four cements (isothermal, $w/c = 0.40$). 66

LIST OF TABLES

Table 3.1: Cement sample identification table	35
Table 3.2: XRD measurement conditions	37
Table 3.3: VCCTL input parameters	54
Table 4.1: Semi-quantitative XRD phase composition of cement samples	61
Table 4.2: Gaussian Model Parameters and Cumulative Heat Release for All Cement Types	65
Table 4.3: VCCTL Simulation Summary: Degree of Hydration and Heat Release at 19h	67
Table 4.4: Hydration Kinetics Correlation	68
Table 5.1: Comparison of experimental and VCCTL cumulative heat with relative difference.	Error! Bookmark not defined.

NOTATION

calorimetric model:

Symbol	Definition	Unit
m_c	Mass of cement	g
m_w	Mass of water	g
m_{total}	Total mass of paste	g
cp_c	Specific heat capacity of cement	J/g – K
cp_w	Specific heat capacity of water	J/g – K
cp_{paste}	Specific heat capacity of cement paste	J/g – K
C_p	Total heat capacity of paste	J/K
C_v	Heat capacity of calorimeter vessel	J/K
C	Total heat capacity of system	J/K
C_a	Adjusted (time – varying) heat capacity	J/K
K_l	Heat loss constant (calibration)	J/min – K
T_s	Sample (cement paste) temperature	°C
T_{amb}	Ambient temperature	°C
ΔT	Temperature difference ($T_s - T_{amb}$)	K
$\frac{dT}{dt}$	Rate of temperature rise	K/min
q	Total heat generation rate	J/min
q_g	Specific heat generation rate	J/min – g
Q	Cumulative heat release	J/g
r	Degree of hydration ratio (relative)	
t	Time elapsed	min

Cement chemistry and XRD:

Symbol	Definition
C_3S	Tricalcium silicate (alite)
C_2S	Dicalcium silicate (belite)
C_3A	Tricalcium aluminate
C_4AF	Tetracalcium aluminoferrite
DOH	Degree of hydration

Modeling:

Symbol	Definition
G_1, G_2	First and second Gaussian components
$a_1 \dots a_6$	Polynomial – 6 fit coefficients
R^2	Coefficient of determination (goodness of fit)
2θ	Diffraction angle (XRD)
d	Interplanar d – spacing

CHAPTER 1

INTRODUCTION

1.1 Background

Cement is the most widely used binding material in the construction industry. It serves as the primary constituent of concrete. The hydration of cement involves chemical reactions between cement particles and water. This process governs the development of strength, durability, and volume stability in concrete. During hydration, heat is released due to exothermic reactions among clinker minerals such as tricalcium silicate (C_3S), dicalcium silicate (C_2S), tricalcium aluminate (C_3A), and tetracalcium aluminoferrite (C_4AF). The magnitude and rate of heat generation significantly affect how concrete temperature changes, especially at early ages.

In mass concrete structures such as dams, bridge foundations, raft foundations, and large retaining walls, the heat generated during cement hydration can lead to substantial temperature rises in the concrete core. The interior concrete is insulated from the environment while the surface cools more rapidly. As a result, temperature gradients develop throughout the structure. These temperature differences induce thermal stresses, which may exceed the tensile capacity of young concrete and result in thermal cracking. Such cracks adversely affect structural durability, serviceability, and performance over time. Therefore, understanding and accurately predicting hydration heat evolution is essential for designing and constructing temperature-sensitive concrete structures.

Hydration characteristics of cement depend on several factors, including chemical composition, mineralogical phases, fineness, water-to-cement ratio, curing conditions, and the presence of supplementary cementitious materials. Ordinary Portland Cement (OPC), commonly classified as CEM I under BDS EN 197-1, generally exhibits rapid hydration and high heat evolution due to its higher clinker content. In contrast, composite cements such as CEM II contain mineral additions including limestone, fly ash, slag, or other supplementary materials that partially replace clinker. These additions modify hydration kinetics, reduce cumulative heat release, and improve thermal performance. As a result, composite cements are increasingly preferred in mass concrete applications where temperature control is critical.

In addition to calorimetric measurements, mineralogical characterization

techniques play a crucial role in explaining hydration behavior. X-ray diffraction (XRD) is widely used to identify and quantify the major clinker phases present in cement. Since the hydration rates of C_3S , C_2S , C_3A , and C_4AF differ significantly, knowledge of phase composition helps explain variations in heat evolution among different cement types. The combination of calorimetric observations and mineralogical analysis therefore provides a comprehensive understanding of hydration mechanisms.

To improve the prediction of hydration behavior, mathematical and computational models have been developed to describe the evolution of heat during cement hydration. Analytical models enable the representation of complex multi-stage hydration processes through simplified mathematical functions, making them suitable for incorporation into thermal simulation software. Furthermore, computational tools such as the Virtual Cement and Concrete Testing Laboratory (VCCTL), developed by the National Institute of Standards and Technology (NIST), allow researchers to simulate hydration processes based on cement composition and microstructural characteristics. Validation of experimental results through such numerical models enhances confidence in hydration predictions and supports their practical application in engineering design.

Although extensive research has been conducted on the hydration behavior of Portland cement and blended cement systems, most studies have focused on cement types commonly used in North America and Europe. Limited information is available on commercially produced Bangladeshi cements that conform to BDS EN 197-1 specifications, particularly regarding their heat evolution characteristics under semi-adiabatic conditions. Furthermore, comparative investigations involving CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N are scarce despite their widespread use in construction projects throughout Bangladesh.

Therefore, this study investigates the hydration behavior of four commercially available cements conforming to BDS EN 197-1:2003, namely CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N. Semi-adiabatic calorimetric measurements are conducted to determine temperature development, heat generation rates, and cumulative heat release. X-ray diffraction analysis is performed to characterize the clinker mineralogy of each cement. The experimental results are subsequently modeled using a two-component Gaussian function and sixth-order polynomial equations to derive continuous analytical representations of the evolution of hydration heat. Finally, the experimental findings are validated using predictions

obtained from the NIST Virtual Cement and Concrete Testing Laboratory (VCCTL). The outcomes of this study are expected to contribute to improved thermal analysis, cement selection, and temperature-control strategies for mass-concrete structures in Bangladesh.

1.2 Significance of the Study

The heat generated during cement hydration plays a crucial role in the early-age performance of concrete structures. Excessive heat evolution can cause significant temperature rises in concrete elements, particularly in mass concrete applications, leading to thermal gradients and thermal stresses. If these stresses exceed the tensile strength of immature concrete, thermal cracking may occur, adversely affecting durability, serviceability, and long-term structural performance.

Accurate characterization of hydration kinetics and heat evolution is therefore essential for predicting temperature development and implementing effective temperature control measures. However, hydration behavior varies considerably among different cement types due to differences in clinker composition, mineral additions, and fineness. Composite cements generally exhibit lower heat evolution than ordinary Portland cement, but the extent of this reduction depends on their specific composition and hydration characteristics.

Although hydration heat models have been developed for various cement systems, limited information is available for commercially produced Bangladeshi cements conforming to BDS EN 197-1 specifications. Furthermore, reliable analytical models suitable for thermal simulations require experimentally validated hydration data under controlled conditions.

This research addresses these challenges by experimentally investigating the hydration behavior of CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N under semi-adiabatic conditions. The study integrates calorimetric measurements, mineralogical characterization, analytical modeling, and numerical simulation to provide a comprehensive understanding of the evolution of hydration heat. The findings are expected to contribute to improved cement selection, thermal analysis, and crack mitigation strategies for mass concrete structures in Bangladesh and similar climatic regions.

1.3 Objectives of the Research

The objectives of this study are as follows:

1. To characterize the mineralogical composition of CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N using X-ray diffraction (XRD) and experimentally determine the temperature development, heat generation rate, and cumulative heat release of the selected cements under semi-adiabatic and sealed hydration conditions.
2. To validate experimental hydration behavior using predictions obtained from the NIST Virtual Cement and Concrete Testing Laboratory (VCCTL) and to provide practical hydration parameters for thermal analysis and temperature control in mass concrete structures.

1.4 Scope of the Study

This study encompasses:

Cement Selection and Characterization of four commercially available cement types conforming to BDS EN 197-1:2003 were selected for this study to represent commonly used Portland and composite cements, namely CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N. The mineralogical characterization of the selected cement samples was performed using X-ray diffraction (XRD) to identify the major crystalline clinker phases. Special attention was given to the identification of the main hydration-contributing compounds, including tricalcium silicate (C_3S), dicalcium silicate (C_2S), tricalcium aluminate (C_3A), and tetracalcium aluminoferrite (C_4AF), as these phases strongly influence the hydration rate, heat evolution, and early-age thermal behavior of cement.

Experimental Investigation of Hydration Behavior of Semi-adiabatic calorimetric testing was conducted under sealed conditions to evaluate the hydration heat behavior of the selected cement samples. During the test, the temperature development of the cement paste was continuously monitored throughout the hydration process. The recorded temperature-time data were then used to determine the heat generation rate and cumulative heat release, which are essential parameters for comparing hydration kinetics and assessing the thermal cracking potential of different cement types.

Hydration Modeling and Analysis:

Analytical hydration models were developed using two-component Gaussian functions to represent the major stages of cement hydration, including the early aluminate reaction and the dominant silicate hydration peak. In addition, the cumulative heat evolution was mathematically represented using sixth-order polynomial equations to obtain smooth continuous functions suitable for thermal analysis. These models were used to evaluate the hydration kinetics and thermal performance of different cement types by comparing peak heat generation rate, peak timing, cumulative heat release, and overall heat evolution behavior.

Numerical Validation:

Cement hydration was simulated using the NIST Virtual Cement and Concrete Testing Laboratory (VCCTL) software to validate the experimentally obtained hydration behavior. The simulated hydration characteristics, including degree of hydration and heat release, were compared with the experimental results obtained from semi-adiabatic calorimetry. This comparison was used to assess the reliability and applicability of the developed analytical models for predicting hydration heat evolution and supporting early-age thermal analysis of cement-based materials.

Comparative Performance Assessment:

The hydration behavior of CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N was compared to evaluate the influence of cement type and mineral addition content on early-age heat evolution. The heat generation rate, cumulative heat release, and peak hydration characteristics were analyzed in relation to thermal cracking potential. Based on these comparisons, suitable cement types were identified for temperature-sensitive and mass concrete applications, where lower heat evolution and slower hydration are preferred to reduce early-age thermal stress and cracking risk.

Engineering Applications:

Practical recommendations for cement selection were developed based on the measured hydration heat behavior, cumulative heat release, and thermal performance of the selected cement types. The experimentally determined hydration parameters, including heat generation rate, cumulative heat release, peak heat rate, and peak timing, can be used as input data for numerical thermal simulations of early-age concrete. These findings also provide guidance for temperature control strategies in mass concrete construction, helping engineers select suitable cement types and adopt proper measures

to reduce excessive temperature rise and thermal cracking risk.

1.5 Organization of Thesis

This thesis has been organized into five chapters.

Chapter 1 presents the background of cement hydration, the significance of hydration heat in concrete structures, the study's objectives, and the scope of the research.

Chapter 2 provides a comprehensive review of the literature related to cement hydration mechanisms, heat evolution, calorimetric techniques, hydration modeling approaches, mineralogical characterization methods, and VCCTL-based simulation studies.

Chapter 3 describes the research methodology, including the materials and cement types used, specimen preparation, semi-adiabatic calorimetry, XRD characterization, data-processing procedures, analytical model development, and VCCTL simulation techniques.

Chapter 4 presents and discusses experimental and numerical results. Comparisons are made among the hydration behaviors of CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N, and the developed hydration models and VCCTL predictions are evaluated and validated.

Chapter 5 summarizes the study's major findings, highlights the practical implications for thermal analysis and mass-concrete construction, identifies the limitations of the research, and provides recommendations for future investigations.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of previous studies related to cement hydration, hydration heat evolution, calorimetric characterization, hydration kinetics modeling, mineralogical analysis, and numerical simulation of cementitious systems. Understanding hydration behavior is essential for predicting temperature development and thermal cracking risks in mass concrete structures. Numerous researchers have investigated the evolution of hydration heat in Portland cement and blended cement systems using experimental, analytical, and computational approaches.

The literature reviewed in this chapter is organized into the following major sections:

- Review of literature on cement hydration mechanisms.
- Review of literature on heat evolution and calorimetric techniques.
- Review of literature on hydration behavior of Portland and composite cements.
- Review of literature on hydration kinetics and analytical modeling.
- Review of literature on X-ray diffraction characterization of cement.
- Review of literature on VCCTL-based hydration simulation.
- Identification of research gaps.

2.2 Cement Hydration Mechanisms

Cement hydration is a complex physicochemical process involving reactions between cement clinker phases and water. The major clinker compounds responsible for hydration are tricalcium silicate (C_3S), dicalcium silicate (C_2S), tricalcium aluminate (C_3A), and tetracalcium aluminoferrite (C_4AF).

Taylor (1997) reported that C_3S is primarily responsible for early-age strength development and contributes significantly to the generation of hydration heat. In contrast, C_2S hydrates more slowly and contributes mainly to long-term strength development.

Bullard et al. (2011) described cement hydration as a sequence of stages including initial reaction, dormant period, acceleration period, deceleration period, and

long-term diffusion-controlled hydration. The heat evolution pattern observed during hydration reflects these stages and provides valuable information regarding reaction kinetics.

The hydration of C_3A occurs rapidly in the presence of gypsum and contributes to the initial exothermic peak observed in calorimetric measurements. Quennoz and Scrivener (2013) showed that gypsum depletion significantly influences aluminate hydration and subsequent heat evolution.

2.3 Heat Evolution During Cement Hydration

Hydration of cement is an exothermic process that releases considerable heat. In mass concrete structures, excessive heat accumulation may cause temperature gradients and thermal cracking.

Klemczak and Batog (2016) investigated hydration heat evolution of low-clinker cements containing slag and fly ash. Their results showed that increasing mineral additive content significantly reduced cumulative heat release compared with ordinary Portland cement.

Pane and Hansen (2005) reported that hydration heat measurements provide reliable information regarding reaction rates and degree of hydration. The authors demonstrated strong relationships between calorimetric observations and microstructural development.

Pang et al. (2013) showed that hydration heat evolution is closely related to chemical shrinkage and hydration progress, demonstrating that isothermal calorimetry can be used as a reliable indicator of cement hydration kinetics.

Several studies have demonstrated that Portland cement generally produces higher heat release than blended cement systems because of its higher clinker content. The magnitude and rate of heat release are further influenced by curing temperature (Han et al., 2014; Zákoutský et al., 2012), and data-driven and analytical models have been proposed to predict the hydration heat of mass concrete (Liu et al., 2021).

2.4 Calorimetric Techniques for Hydration Analysis

Calorimetry is one of the most widely used techniques for evaluating hydration kinetics.

2.4.1 Isothermal Calorimetry

Isothermal calorimetry measures heat flow while maintaining a constant temperature. Schindler and Folliard (2005) used isothermal calorimetry to investigate hydration characteristics of Portland cement blended with supplementary cementitious materials, and large isothermal-calorimetry databases now provide analytical approximations of heat-release curves for many cements (Šmilauer et al., 2026).

Although isothermal calorimetry provides accurate heat-flow measurements, it does not fully represent field conditions in which concrete temperature changes continuously.

2.4.2 Semi-Adiabatic Calorimetry

Semi-adiabatic calorimetry allows limited heat exchange with the surroundings and more closely simulates practical concrete placements.

Klemczak and Batog (2016) demonstrated that semi-adiabatic measurements provide realistic temperature development profiles suitable for thermal analysis of mass concrete structures; standardized adiabatic and semi-adiabatic procedures for this purpose are described by the RILEM Technical Committee 119-TCE (1997), and Hernandez-Bautista et al. (2016) numerically reproduced semi-adiabatic temperature development in cement mortar.

Recent studies have explored the use of semi-adiabatic calorimetry to evaluate setting times, hydration rates, and temperature-rise characteristics of cementitious materials. However, limited information is available on commercially available Bangladeshi cements.

2.4.3 Hydration Behavior of Portland and Composite Cements

Ordinary Portland cement generally exhibits rapid hydration and high heat evolution because of its high clinker content.

Composite cements contain supplementary materials such as fly ash, slag, limestone, or other mineral constituents. These materials modify hydration kinetics and reduce cumulative heat release.

Lothenbach et al. (2011) reported that slag-containing cements exhibit delayed hydration but improved long-term performance. Snelson et al. (2008) found that fly ash significantly reduces peak heat evolution and lowers the risk of thermal cracking, and

Han et al. (2019) reported similar refinements in hydration for ultrafine fly ash–cement composites.

Research by Klemczak and Batog (2016) showed that increasing slag or fly ash replacement reduces cumulative heat generation and delays peak hydration rates.

Although numerous studies have investigated binary cement systems, relatively few studies have focused on commercially manufactured CEM II A-M and CEM II B-M cements containing mixed mineral constituents.

2.5 Hydration Kinetics and Analytical Modeling

De Schutter and Taerwe (1995) proposed a degree of hydration-based model that expresses the heat generation rate as a function of the normalized degree of hydration, capturing the characteristic bell-shaped hydration curve observed in calorimetric measurements. Their formulation has been widely adopted in thermal analysis of mass concrete due to its simplicity and reasonable predictive accuracy.

Mills (1966) established an empirical relationship between the ultimate degree of hydration and the water-to-cement ratio, providing a foundational expression for estimating the maximum extent of cement hydration under sealed curing conditions. Rastogi et al. (2024) demonstrated that the NIST Virtual Cement and Concrete Testing Laboratory (VCCTL) could reliably simulate cement hydration and heat release by incorporating mineralogical composition from XRD analysis, enabling virtual experimentation as a complement to laboratory calorimetry. Multi-phase kinetic models extending this framework to blended cements have since been developed (Meinhard & Lackner, 2008; Wang et al., 2010; Wang, 2014).

Lothenbach et al. (2011) highlighted the importance of supplementary cementitious materials in modifying hydration kinetics and thermodynamic phase assemblages, emphasizing that composite cement systems require phase-specific modeling approaches. Despite these advances, the direct integration of XRD-derived clinker phase quantification with semi-adiabatic calorimetric data for calibrating Gaussian-based heat evolution models remains insufficiently explored for composite cements conforming to BDS EN 197-1:2003. The present study addresses this gap by combining mineralogical characterization with analytical Gaussian and polynomial hydration models, validated against VCCTL simulations.

2.6 X-Ray Diffraction Characterization of Cement

X-ray diffraction (XRD) is widely used to characterize cement mineralogy and hydration products.

Scrivener et al. (2015) reported that XRD enables identification of clinker phases including C_3S , C_2S , C_3A , and C_4AF . Quantification of these phases helps explain variations in hydration behavior among different cement types.

Several studies have combined XRD with calorimetric measurements to correlate mineralogical composition with hydration heat evolution. However, such integrated studies remain limited for commercially available CEM II composite cements.

2.7 Numerical Simulation Using VCCTL

The Virtual Cement and Concrete Testing Laboratory (VCCTL), developed by NIST, is a powerful computational tool for simulating hydration and microstructure development.

Ferraro et al. (2018) demonstrated that VCCTL accurately predicts degree of hydration, chemical shrinkage, and heat evolution in ordinary Portland cement systems.

Recent studies have extended VCCTL applications to blended cement systems containing fly ash and slag. Validation results generally show good agreement between simulated and experimental hydration characteristics.

Despite these advances, validation studies involving commercially available composite cements containing mixed mineral additions remain relatively scarce.

2.8 Research Gaps

Based on the reviewed literature, several research gaps have been identified regarding the hydration behavior of Portland and composite cements. Limited studies are available on commercially produced Bangladeshi cements conforming to BDS EN 197-1, particularly for commonly used cement types such as CEM I 52.5 N, CEM II A-M 42.5 N, and CEM II B-M 42.5 N. The comparative hydration behavior of these cements has not been systematically investigated under semi-adiabatic conditions. Most previous studies have mainly relied on isothermal calorimetry, which does not fully represent the partial heat-retention condition commonly observed in mass concrete

structures.

Moreover, the integration of XRD characterization, calorimetric measurements, analytical modeling, and VCCTL validation within a single research framework remains limited. The use of two-component Gaussian models to separately represent aluminate and silicate hydration peaks has received limited attention, and continuous polynomial representations of cumulative heat evolution suitable for thermal simulations are rarely reported. In addition, VCCTL validation for composite cements containing mixed secondary constituents is still insufficiently documented, and practical hydration parameters for thermal analysis of mass concrete using locally available cements are not readily available.

Therefore, the present study aims to address these research gaps through experimental characterization, analytical modeling, and numerical validation of hydration behavior in Portland and composite cements under semi-adiabatic conditions.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter presents the methodology adopted to investigate the hydration behavior of Portland and composite cements under semi-adiabatic conditions. The study was conducted on commercially available cements conforming to BDS EN 197-1:2003, namely CEM I 52.5 N, CEM II A-M 42.5 R, CEM II A-M 42.5 N and CEM II B-M 42.5 N. The main purpose of the methodology was to experimentally determine the evolution of hydration heat, develop analytical models of heat evolution, and validate the results using numerical simulations.

The methodology consists of four major components: material selection and sample preparation, mineralogical characterization using X-ray diffraction (XRD), semi-adiabatic calorimetric testing, and analytical/numerical modeling. XRD analysis was used to identify the major mineral phases present in the selected cements. Semi-adiabatic calorimetry was then conducted under sealed conditions to record temperature development during hydration. The recorded temperature data were processed in MATLAB to calculate heat generation rate and cumulative heat release. The heat generation curves & cumulative heat curves were represented using two-component Gaussian model & sixth-order polynomial equations. Finally, the experimental hydration behavior was compared with predictions obtained from the NIST Virtual Cement and Concrete Testing Laboratory (VCCTL).

The complete methodology was designed to provide a systematic comparison of the thermal behavior of Portland and composite cements and to generate practical input parameters for early-age thermal analysis of mass concrete structure. The overall research framework is illustrated schematically in Figure 3.1.

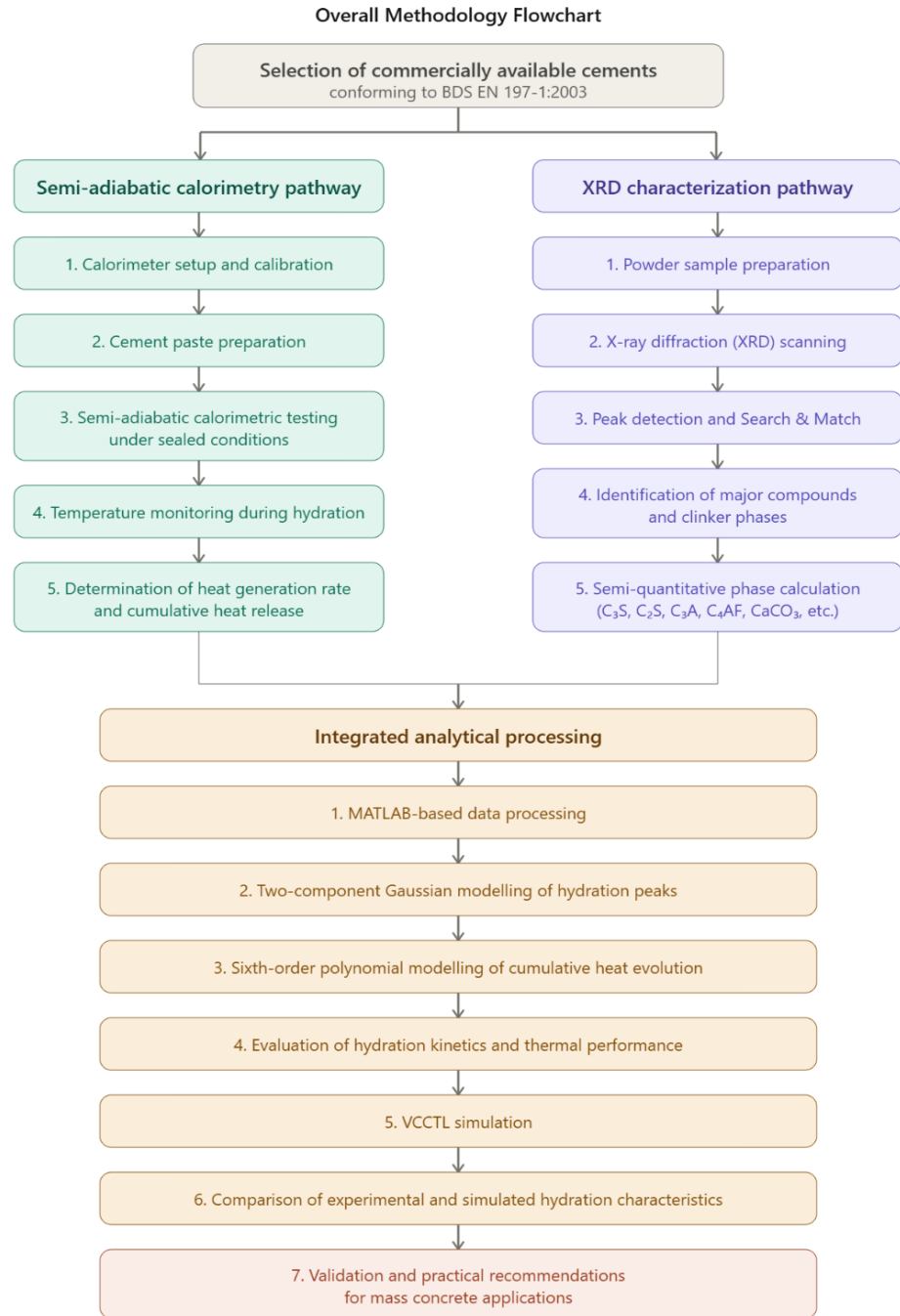


Figure 3.1: Overall Methodology Flowchart

3.2 Methodology Overview

The overall research procedure followed a sequential experimental and

analytical framework. First, the cement samples were collected and stored under controlled conditions to prevent moisture absorption. The mineralogical phases of the samples were then identified using XRD analysis. After that, cement paste specimens were prepared at selected water-to-cement ratios and tested in a semi-adiabatic calorimeter. The measured temperature-time data were converted into heat generation rate and cumulative heat release using a heat balance equation.

After processing the experimental data, the hydration heat generation rate curves were fitted with a two-component Gaussian model. This model was selected because cement hydration commonly exhibits multiple exothermic peaks, primarily associated with aluminate and silicate reactions. In addition, cumulative heat release was fitted using sixth-order polynomial equations. The detailed research workflow is presented in Figure 3.2.

Sixth-order polynomial equations were used to obtain smooth, continuous expressions suitable for thermal simulation. Finally, VCCTL simulations were performed using XRD-derived cement phase information, and the simulation outputs were compared with the experimental curves.

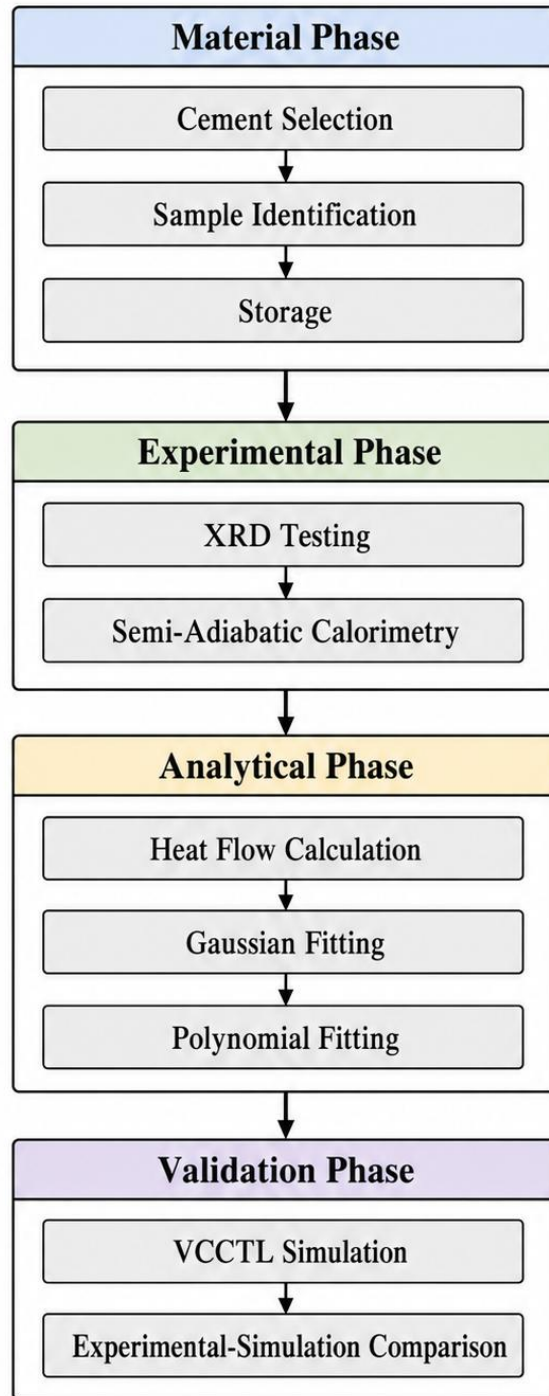


Figure 3.2: Detailed Research Workflow Flowchart

3.3 Materials Used

Four commercially available cement types were selected for this study. These cement types were chosen because they are widely used in concrete construction and represent different clinker and mineral addition contents.

The selected cement types were:

CEM I 52.5 N:

CEM I 52.5 N is an Ordinary Portland Cement with high clinker content. Because of its higher clinker percentage, it is expected to show rapid hydration, high early-age heat generation, and higher cumulative heat release. This cement was included to represent a high-heat Portland cement system.

CEM II A-M 42.5 R:

CEM II A-M 42.5 R is a Portland composite cement containing a moderate proportion of mineral additions, like CEM II A-M 42.5 N, but with a higher early-strength development (Rapid hardening, "R" class). Enhanced clinker reactivity and finer grinding promote faster hydration in the early stages, resulting in a higher initial heat evolution rate than the corresponding 42.5 N grade. However, due to the partial replacement of clinker by supplementary constituents, the overall heat release remains lower than that of CEM I cement. Consequently, CEM II A-M 42.5 R provides a balance between rapid strength gain and moderate heat of hydration, making it suitable for applications requiring early-age performance while reducing the risk of excessive temperature rise.

CEM II A-M 42.5 N:

CEM II A-M 42.5 N is a Portland composite cement containing a moderate number of mineral additions. The partial replacement of clinkers by supplementary constituents is expected to reduce the rate and amount of hydration heat compared with CEM I

CEM II B-M 42.5 N:

CEM II B-M 42.5 N is a Portland composite cement with a higher proportion of mineral additions than CEM II A-M. This cement is expected to show slower hydration and lower cumulative heat release, making it potentially more suitable for mass concrete or temperature-sensitive concrete applications.

All cement samples were collected from sealed bags and stored in airtight containers before testing. The storage procedure was maintained to reduce the possibility of pre-hydration, moisture absorption, and carbonation. Photographs of the four cement samples used in the experimental program are shown in Figure 3.3.

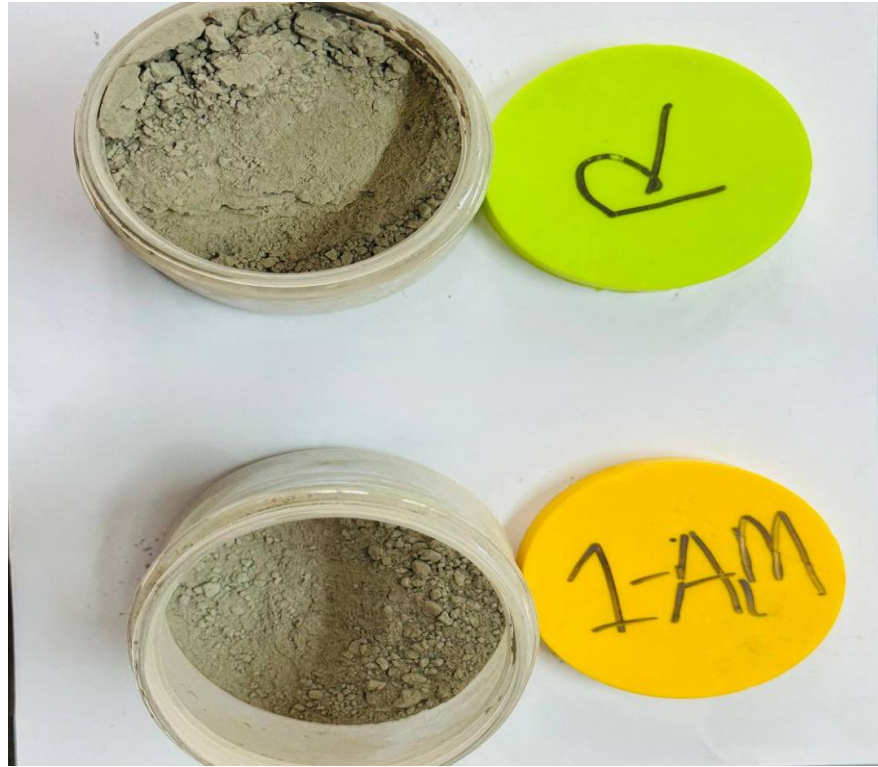


Figure 3.3: Cement samples used in the experimental program

3.4 Sample Identification

For laboratory traceability, the XRD-tested cement samples were identified as Cement 1, Cement 2, Cement 3, and Cement 4. The XRD data were used to identify the major crystalline phases present in the cement samples and to support the interpretation of hydration behavior.

Although the main thesis studies four cement types, the XRD dataset is used to support the comparison and verification of mineralogical variation among all tested samples. The correspondence between the commercial cement type and the XRD sample ID is summarized in Table 3.1.

Table 3.1: Cement sample identification table

Sample	Cement Type	Test Performed	Remarks
Cement 1	CEM II B-M 42.5 N	XRD / calorimetry	Used for phase identification
Cement 2	CEM II A-M 42.5 N	XRD / calorimetry	Used for phase identification
Cement 3	CEM II A-M 42.5 R	XRD / calorimetry	Used for phase identification
Cement 4	CEM I 52.5 N	XRD / calorimetry	Used for phase identification

3.5 Cement Paste Preparation

Cement paste specimens were prepared using the selected cement samples and clean mixing water. The water-to-cement ratio was maintained at 0.40 and 0.50 depending on the specific calorimetry trial. The same mixing procedure was followed for all cement types to ensure comparability of hydration results.

The paste preparation procedure was as follows:

For each calorimetry test, the required amount of cement was first weighed accurately, and the corresponding amount of water was measured according to the selected water-to-cement ratio. Water was placed into the mixing container first, after which cement was gradually added to minimize lump formation. The paste was mixed manually for 90 seconds, followed by mechanical mixing at low speed for 60 seconds to obtain a uniform and homogeneous paste. Immediately after mixing, the fresh cement paste was transferred into the calorimeter vial, and the vial was sealed properly to prevent evaporation and moisture exchange during the test. The time of first contact between cement and water was recorded as ($t = 0$). A fixed paste volume of approximately 200 mL was used for each semi-adiabatic calorimetry test.

All specimens were prepared using the same procedure to minimize experimental variation. The preparation of cement paste specimens for semi-adiabatic calorimetry is illustrated in Figure 3.4.



Figure 3.4: Preparation of cement paste specimen for semi-adiabatic Calorimetry

3.6 Mineralogical Characterization by X-Ray Diffraction

3.6.1 Purpose of XRD Analysis

X-ray diffraction analysis was conducted to identify the crystalline mineral phases present in the cement samples. Cement hydration heat depends strongly on the relative proportions of clinker phases such as alite, belite, aluminate, and ferrite phases. Therefore, XRD analysis was used to support the explanation of the calorimetric results.

The main phases considered in this study were:

Alite or tricalcium silicate-related phase, C_3S

Belite or dicalcium silicate-related phase, C_2S

Tricalcium aluminate-related phase, C_3A

Calcium aluminoferrite-related phase, C_4AF

Calcium carbonate, $CaCO_3$

Quartz or silicon dioxide, SiO_2

Calcium-magnesium-aluminum silicate-type phases

Alite is mainly responsible for early strength development and the dominant silicate hydration peak. Belite reacts more slowly and contributes to later-age hydration. Aluminate phases react rapidly in the presence of gypsum, contributing to the early heat-evolution peak. Calcium carbonate and quartz are comparatively less reactive and may act mainly as filler phases.

3.6.2 XRD Instrumentation and Test Conditions

The XRD test was conducted using a PANalytical EMPYREAN diffractometer, as shown in Figure 3.5. The instrument was equipped with a copper radiation source. The test was performed using a flat sample stage, and the powdered cement samples were scanned over a wide 2θ range to identify the major crystalline phases. The detailed measurement conditions are summarized in Table 3.2.



Figure 3.5: X-ray diffraction equipment used for mineralogical characterization.

Table 3.2: XRD measurement conditions

Parameter	Value
Diffractometer	PANalytical EMPYREAN
Radiation source	Cu K α
K α 1 wavelength	1.54060 Å
Scan range	6.0056°–89.9856° 2θ
Step size	0.0260° 2θ
Scan step time	7.14 s

Scan type	Continuous
PSD mode	Scanning
PSD length	3.35° 2 θ
Divergence slit type	Fixed
Divergence slit size	0.7197°
Specimen length	10 mm
Measurement temperature	25°C
Generator setting	40 mA, 45 kV
Sample spinning	No

Powder samples were placed in flat sample holders before scanning, as illustrated in Figure 3.6 The obtained diffraction patterns were later analyzed to identify crystalline compounds.



Figure 3.6: Preparation of powdered cement sample for XRD analysis

3.6.3 Phase Identification and Semi-Quantitative Estimation

The XRD test was conducted using a PANalytical EMPYREAN diffractometer. The instrument was equipped with a copper radiation source. The test was performed using a flat sample stage, and the powdered cement samples were scanned over a wide 2 θ range to identify the major crystalline phases.

The XRD patterns were analyzed using Search and Match procedures. The measured peak positions and relative intensities were compared with reference diffraction patterns to identify the possible crystalline phases present in the cement

samples.

Semi-quantitative phase estimation was performed using the Reference Intensity Ratio (or scale factor) method. The relative phase percentage was calculated using:

$$\text{Phase percentage of compound, } i = \frac{\text{Scale factor of compound } i}{\text{Sum of all scale factors}} \times 100$$

This method provides approximate relative phase proportions rather than exact absolute phase quantities. Therefore, the XRD results were interpreted as semi-quantitative values. The phase contents were used to explain differences in hydration behavior and to prepare approximate input parameters for VCCTL simulation. The complete XRD analysis workflow is presented in Figure 3.7.

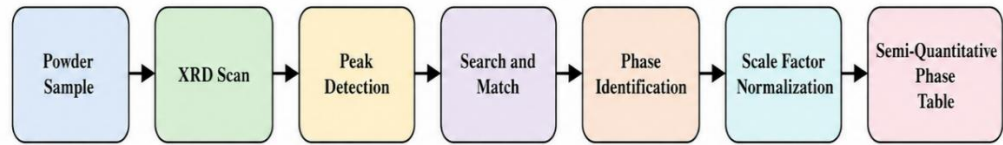


Figure 3.7: XRD analysis workflow chart

3.7 Semi-Adiabatic Calorimetry

3.7.1 Purpose of Semi-Adiabatic Calorimetry

Semi-adiabatic calorimetry was performed to monitor the temperature rise caused by cement hydration under sealed conditions. This method was selected because it more realistically simulates the heat-retaining behavior of mass concrete than fully isothermal testing. In mass concrete, the heat of hydration is not dissipated immediately, and an internal temperature rise may lead to thermal stress and cracking.

The objectives of the calorimetry test were:

The semi-adiabatic calorimetry test was conducted to record the temperature development of cement paste during hydration and to evaluate the heat evolution characteristics of the selected cement samples. The recorded temperature-time data were used to determine the heat generation rate, cumulative heat release, and peak hydration time. These parameters were then used to compare the hydration kinetics of CEM I and CEM II cements and to assess their relative thermal cracking potential in temperature-sensitive and mass concrete applications.

3.7.2 Semi-Adiabatic Calorimeter Setup

A custom-built bench-top semi-adiabatic calorimeter was used in this study. The calorimeter consisted of an insulated outer container and a sealed inner vial containing the cement paste specimen. The insulation reduced heat loss to the surroundings, while the sealed vial prevented moisture exchange during hydration.

The setup consisted of:

Insulated outer container

Inner sample vial

Cement paste specimen

Sample temperature sensor

Ambient temperature sensor

Data logger

Computer for data storage

Temperature was recorded at 60-second intervals. Both the sample temperature and the ambient temperature were recorded to enable heat-loss correction during data processing. The semi-adiabatic calorimeter used in this study is shown in Figure 3.8, and a schematic diagram of the full calorimeter setup is presented in Figure 3.9.



Figure 3.8: Semi-adiabatic calorimeter used for hydration heat measurement

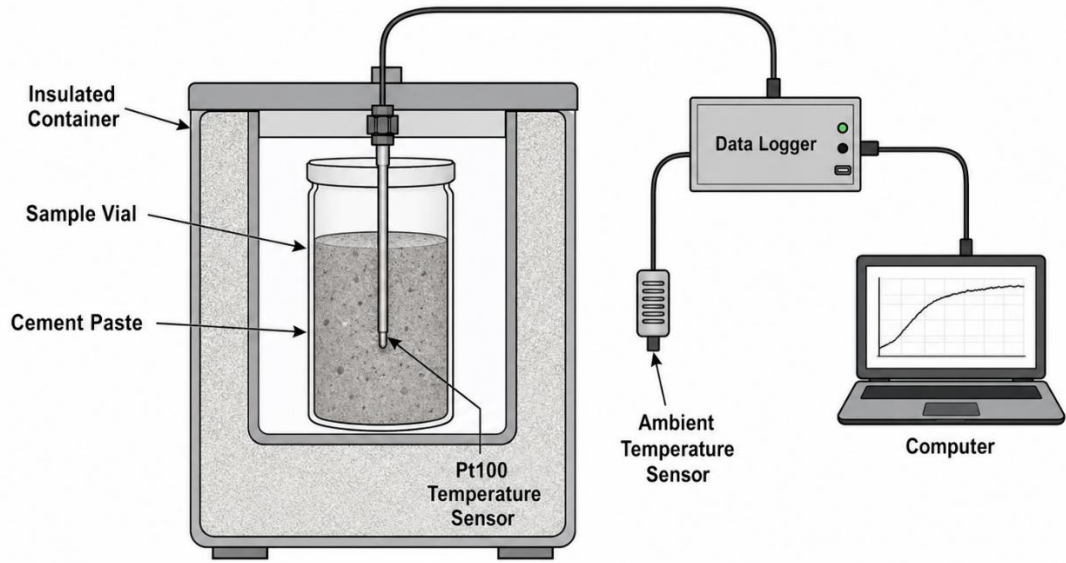


Figure 3.9: Schematic diagram of calorimeter setup

3.7.3 Calibration of the Calorimeter

Before conducting hydration tests, the calorimeter was calibrated to determine two important parameters: effective heat capacity and heat-loss coefficient. These parameters were required to convert measured temperature rise into heat generation rate.

Electrical Calibration

Electrical calibration was conducted to determine the calorimeter system's effective heat capacity. A resistor heater with known power was placed inside the calorimeter vial. The heater was operated for a known duration, and the resulting temperature rise was measured.

The effective heat capacity was calculated as:

$$C = \frac{P * \tau}{\Delta T} \quad 3.1$$

where:

C = effective heat capacity of the calorimeter system, J/K

P = heater power, W

τ = heating duration, s

ΔT = measured temperature rise, K

Mass $\times$ specific heat (approximate): $C \approx m_{sample}c_{sample} + C_{vessel}$, using a literature value of c for the sample heat capacity.

Cooling Curve Calibration

Cooling calibration was performed to determine the heat-loss coefficient. The vial was heated above ambient temperature and then allowed to cool naturally. The temperature decay was assumed to follow an exponential cooling relationship:

[Heat the vial a few °C above ambient, then record $T(t)$ as it cools naturally to ambient (no reaction):

$$\frac{dT}{dt} = -\frac{K_l}{C}(T - T_{amb}) \quad 3.2$$

By Integrating, 3.3

$$(T(t) - T_{amb}) = (T_0 - T_{amb})e^{-\frac{K_l}{C}t}$$

Fit an exponential to the measured cooling curve to get $\alpha = \frac{K_l}{C}$.

Then, $K_l = \alpha C$

where:

T_t = sample temperature at time t

T_{amb} = ambient temperature

T_0 = initial temperature

K_l = heat – loss coefficient, W/K

C = effective heat capacity, J/K

The heat-loss coefficient was determined by fitting the cooling curve, as described by Equations 3.2 and 3.3. The complete calibration procedure is summarized in the flowchart shown in Figure 3.10.

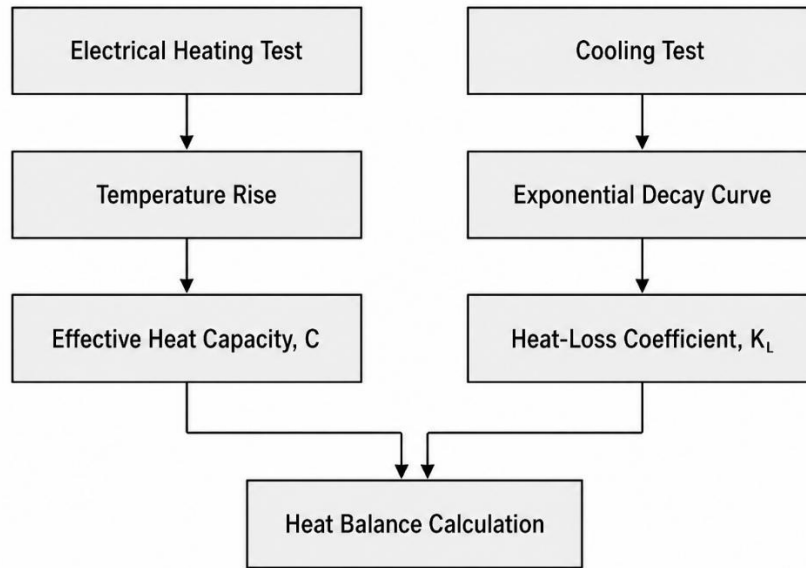


Figure 3.10: Calorimeter calibration flowchart

3.7.4 Hydration Test Procedure

The hydration test was performed using the same procedure for all cement types. After preparing the cement paste, the fresh paste was immediately transferred into the calorimeter vial. The vial was sealed to prevent moisture loss, and the temperature sensor was placed at the center of the paste specimen. The vial was then inserted into the insulated calorimeter.

The test procedure was as follows:

For each hydration test, cement and water were weighed according to the selected water-to-cement ratio, which was approximately 0.41. The cement paste was prepared by following the standard mixing procedure to ensure uniform consistency. Immediately after mixing, the fresh paste was placed inside the inner calorimeter vial, and the vial was sealed to prevent moisture loss and evaporation. The temperature sensor was then carefully inserted into the paste, and the sealed vial was placed inside the insulated calorimeter. During the test, both sample temperature and ambient temperature were recorded at 60-second intervals. Data logging continued throughout the hydration period to capture complete temperature development. After completion of the test, the recorded temperature data were exported to Excel and later analyzed using MATLAB to determine heat generation rate, cumulative heat release, and

hydration behavior. The temperature monitoring and data acquisition system during a typical hydration test is illustrated in Figure 3.11.

Apparatus Description:

The semi-adiabatic calorimetry setup consisted of several basic laboratory apparatus used for calibration, temperature monitoring, and heat-loss control. A 500 W, 220 V electric heater was used during the calibration stage to supply a known amount of heat to the calorimeter system, which helped in determining the effective heat capacity of the setup. An HTC-2 digital LCD temperature-humidity meter was used to monitor the ambient room temperature and relative humidity during the experiment, ensuring that environmental conditions were recorded throughout the test. A DT 9205A multimeter was used to measure electrical parameters such as voltage, current, and resistance during heater calibration. An insulated container was used as the main calorimeter body to reduce heat exchange with the surrounding environment and maintain semi-adiabatic conditions. The insulator helped minimize external heat loss so that the measured temperature rise could be related more accurately to the heat generated during cement hydration.



Figure 3.11: Temperature monitoring and data acquisition during semi-adiabatic calorimetry

3.8 Heat Flow Calculation

3.8.1 Governing Heat Balance Equation

Under semi-adiabatic conditions, the heat generated by cement hydration is partly stored inside the calorimeter system and partly lost to the surrounding environment. The heat balance equation can be expressed as:

The reaction heat flow is inferred from the temperature change and heat

losses.

lumped heat capacity –

$$C \frac{dT}{dt} = \dot{q}_{reaction}(t) - K_l(T(t) - T_{amb}(t))$$

where,

$C = mc = \text{effective capacity of sample (J/k)}$

$K_l = \text{heat – loss coefficient (W/K)}$

$\dot{q}_{reaction} = \text{instantaneous heat flow from reaction (W)}$

Rearrange the above equation to get instantaneous heat flow,

$$\dot{q}_{reaction}(t) = C \frac{dT}{dt} + K_l(T(t) - T_{amb}(t)) \quad 3.4$$

By integrating, we will find the total heat from time elapsing 0 to t1:

$$\begin{aligned} Q &= \int_0^{t_1} \dot{q}_{reaction}(t) dt \\ &= C \int_0^{t_1} dT + \int_0^{t_1} K_l(T(t) - T_{amb}(t)) dt \end{aligned} \quad 3.5$$

where:

$\dot{q}_{reaction}(t) = \text{instantaneous heat generation rate, W}$

$C = \text{effective heat capacity of the system, } \frac{J}{K}$

$\frac{dT}{dt} = \text{rate of temperature change, K/s}$

$K_l = \text{heat – loss coefficient, W/K}$

$T(t) = \text{sample temperature, } ^\circ\text{C}$

$T_{amb}(t) = \text{ambient temperature, } ^\circ\text{C}$

The first term, $C(dT/dt)$, represents the heat stored in the calorimeter system. The second term, $K_l[T(t) - T_{amb}(t)]$, represents heat lost to the surroundings. Therefore, the total heat generated by hydration was obtained by summing these two terms, as expressed by Equations 3.4 and 3.5. The heat calculation flowchart is shown

in Figure 3.12.

3.8.2 Heat Generation Rate per Unit Mass

The instantaneous heat generation rate was converted to heat generation rate per unit mass of cement using:

$$q(t) = \frac{\dot{q}_{reaction}(t)}{m}$$

where:

$q(t)$ = heat generation rate per unit mass of cement

m = mass of cement used in the paste specimen

The final heat generation rate was expressed in J – min/g.

3.8.3 Cumulative Heat Release

Cumulative heat release was calculated by numerical integration of the heat generation rate curve using Simpson's 1/3 Rule, as given by Equations 3.6 and 3.7:

$$Q(t) = \int_0^t q(t)dt$$

Simpson's method of numerical integration has been used to quantify heat release:

Simpson's 1/3 Rule

Divide the time interval into **n even subintervals** (n must be even):

$$\text{Step size: } h = \frac{t_n - t_0}{n}; n = 2 \quad 3.6$$

Discrete values: $q_0, q_1, q_2, \dots, q_n$

$$\text{Formula: } Q \approx \frac{h}{3} [q_0 + 4(q_1 + q_3 + \dots + q_{n-1}) + 2(q_2 + q_4 + \dots + q_{n-2}) + q_n] \quad 3.7$$

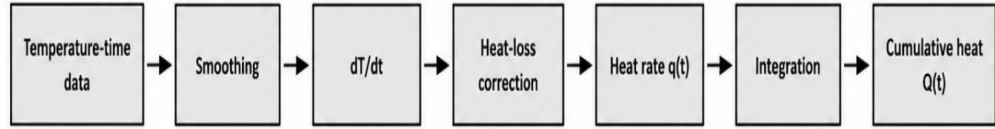


Figure 3.12: Heat calculation flowchart

3.9 Degree of Hydration and Variable Heat Capacity

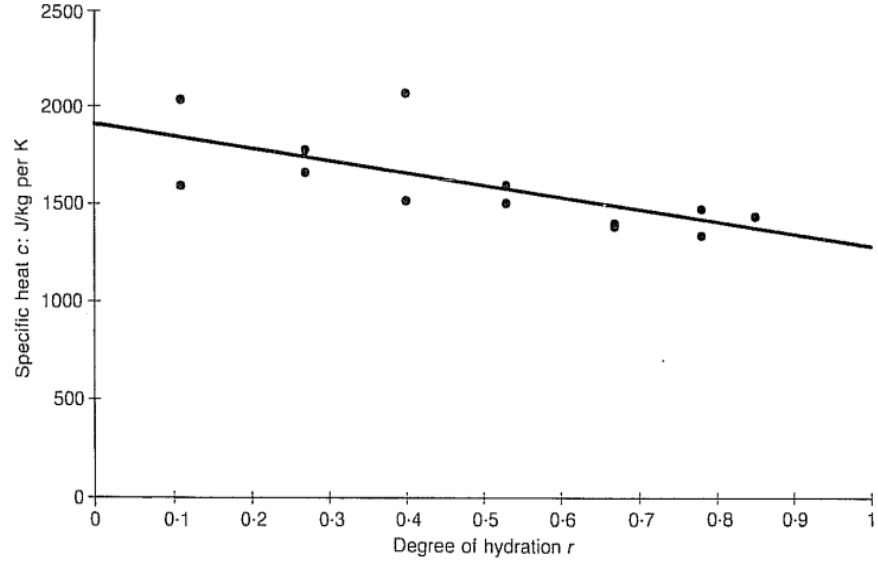


Figure 3.13: Specific heat of hardening cement paste

De Schutter and Taerwe (1995b) established a relationship between the specific heat of cement paste and the degree of hydration, as illustrated in Figure 3.13. For CEM III/C 32.5 (slag rich cement) with w/c ratio 0.5, the derived relation is given by Equations 3.8 and 3.9.

$$C_p(r) = 1300(1.5 - 0.5r) \text{ J/kg per kelvin} \quad 3.8$$

$$3.9$$

$$\text{or } C_p(r) = 1.3(1.5 - 0.5r) \text{ J/kg per kelvin}$$

Here, we use a w/c ratio of 0.4; our cement material is CEM II AM 42.5R. Basically, the heat capacity of cement decreases as the degree of hydration increases, leading to changes in the microstructure.

Studies on hardening cement paste and concrete report a reduction in specific heat capacity as hydration advances (Bentz, 2007; Wolterbeek & Hangx, 2023). CEM II A-M 42.5R might show a similar linear decline as the degree of hydration increases

from 0 to 1. This aligns with Portland systems due to similarities in microstructural formation, though slag cements may have slightly lower final heat capacity due to denser microstructures.

For RHC (pure Portland), the initial heat capacity might be comparable, but the drop occurs faster due to faster increase in degree of hydration. But the early strength cement used has not been manufactured by adding slag or other minerals; the early strength is achieved through the fineness of the cement powder and other minor ingredients.

Based on linear property, we may use the linear relationship as like

$$C_p(r) = C_p(0) + rk \quad 3.10$$

However, we have used the linear property and modelled an approximate equation for our experimental result, as given by Equation 3.11:

$$C_p = 1160.67 * (1.5 - 0.5 * r) \quad 3.11$$

where,

$$C_p(r = 0) \approx m_{sample}c_{sample} + C_{vessel}$$

$$m_c = 139 \text{ g}$$

$$m_w = 57 \text{ g}$$

$$m_{total} = m_c + m_w ; m_c \text{ \& } m_w \text{ indicates } \\ \text{cement and water weight consecutively}$$

$$cp_c = 0.84 \text{ J/g/K ; typical value for calculation}$$

$$cp_w = 4.18 \text{ J/g/K ; typical value for calculation}$$

$$C_{p_{paste}} = \frac{(cp_c * m_c + cp_w * m_w)}{m_{total}} \quad 3.12$$

$$C_p = C_{p_{paste}} * m_{total} \text{ in J/K} \quad 3.13$$

$$C_v = 1386.6 \text{ J/k ; } C_v \text{ due to the heat capacity of vessel}$$

$$C = C_p + C_v \text{ J/K ; total heat capacity of system}$$

Here, the 'r' indicates the ultimate degree of hydration, the constant 'k' depends on the type and ingredients of cement powder.

Degree of hydration [r]

According to De Schutter G. et al. based prediction of thermal characteristics of hardening concrete; 1992 –

The degree of hydration is defined by $r(t)$ of cementitious material defined as the fraction of cement that has already hydrated:

$$r(t) = \frac{\text{amount of cement that has reacted at time } t}{\text{total amount of time at } t = 0} \quad 3.14$$

A practical method to estimate the degree of hydration is based on the exothermic character of the hydration process:

$$r(t) = \frac{Q(t)}{Q_{max}} \quad 3.15$$

Where $Q(t)$ is the total heat developed at time t (in J/g), and Q_{max} is the total heat development corresponding to complete hydration (in J/g).

The total heat $Q(t)$ can be calculated as

$$Q(t) = \int_0^t q(t)dt \quad 3.16$$

Where $q(t)$ is the heat production (in J/g per h)

In ref. 2 a new hydration model is developed, yielding a heat production q depending on the actual degree of hydration $r(t)$ and the actual temperature $\theta(t)$

$$q(t) = q[r(t), \theta(t)] \quad 3.17$$

From the third consecutive equation above, we can establish the following relationships:

$$\frac{dQ(t)}{dt} = q(t) = q[r(t), \theta(t)] \quad 3.18$$

Now, differentiating the second consecutive equation and the introduction of equation yield,

$$dr(t) = \frac{dQ(t)}{Q_{max}} = \frac{q[r(t), \theta(t)]}{Q_{max}} dt \quad 3.19$$

Hence, the value of r can be updated by analyzing the above equation 3.14 to 3.19 which ultimately results in 3.9.14,

$$r(t + dt) = r(t) + dr(t) \text{ or} \quad 3.20$$

$$r(t + \Delta t) = r(t) + \Delta r \quad 3.21$$

Note: During the quantification of ‘ r ’ we have used VCCTL simulated heat generation rate data.

3.10 MATLAB-Based Data Processing

The temperature-time data recorded by the data logger were exported to Excel and imported into MATLAB. Before the heat calculation, preprocessing was performed to improve data reliability.

The preprocessing steps were:

The recorded temperature data were processed systematically before calculating hydration heat parameters. First, the sample temperature and ambient temperature data were imported from the recorded dataset, and the corresponding time values were converted into hours. All data were aligned by taking the time of first contact between cement and water as ($t = 0$). Abnormal, incomplete, or inconsistent readings were removed to improve data quality, and smoothing was applied to reduce noise in the temperature-time curve. The rate of temperature change (dT/dt) was then calculated using numerical differentiation. After that, heat-loss correction was applied using the calibrated heat-loss coefficient (K_1). Finally, the corrected heat generation rate was calculated, and the heat generation rate curve was integrated to obtain the cumulative

heat release during cement hydration. The complete MATLAB data processing workflow is illustrated in Figure 3.14.

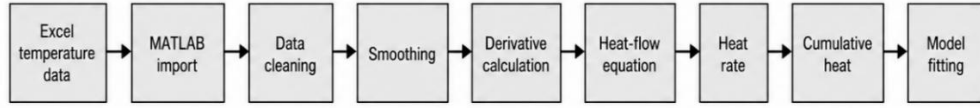


Figure 3.14: MATLAB data processing workflow

3.10.2 Output of Data Processing

The MATLAB analysis produced the following outputs for each cement type:

Temperature rise versus time

Heat generation rate versus time

Cumulative heat release versus time

Gaussian fitted heat generation curve

Polynomial fitted cumulative heat curve

Peak heat rate

Time of peak heat generation

Final cumulative heat release

3.10.3 Two -Component Gaussian Model

The heat generation rate curve for cement hydration typically consists of multiple exothermic peaks. The early peak is mainly associated with aluminate reaction, while the major peak is associated with silicate hydration. To separate and represent these overlapping hydration stages, a two-component Gaussian model was used.

The heat generation rate was expressed as a sum of two Gaussian functions (Equation 3.22):

$$q(t) = A_1 e^{\left\{-\left(\frac{t-\mu_1}{\sigma_1}\right)^2\right\}} + A_2 e^{\left\{-\left(\frac{t-\mu_2}{\sigma_2}\right)^2\right\}} \quad 3.22$$

where:

A_1 = amplitude of the first peak

μ_1 = time of the first peak

σ_1 = width parameter of the first peak

A_2 = amplitude of the second peak

μ_2 = time of the second peak

σ_2 = width parameter of the second peak

The first Gaussian component represents the early aluminate-related reaction, while the second Gaussian component represents the dominant silicate hydration peak.

The fitting was performed in MATLAB using a non-linear least-squares approach. Initial peak locations were selected based on the experimental heat generation curves. Parameter bounds were applied to prevent physically unrealistic fitting results.

The fitting quality was assessed using:

Coefficient of determination, R^2

Root mean square error, RMSE

Agreement between measured and fitted peak time

Agreement between measured and fitted peak amplitude

Visual comparison between measured and fitted curves. The Gaussian fitting methodology is illustrated schematically in Figure 3.15.

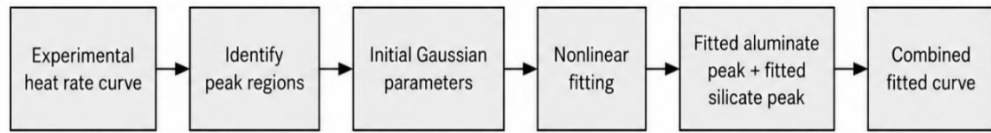


Figure 3.15: Gaussian fitting methodology chart

3.10.4 Polynomial Fitting of Cumulative Heat

To obtain continuous analytical expressions for cumulative heat release, the cumulative heat curves were fitted using a sixth-order polynomial.

The polynomial equation was expressed as Equation 3.10.2:

$$Q(t) = c_0 + c_1t + c_2t^2 + c_3t^3 + c_4t^4 + c_5t^5 + c_6t^6 \quad 3.23$$

where:

$Q(t)$ = cumulative heat release at time t

t = hydration time

c_0 to c_6 = polynomial coefficients

The polynomial coefficients were determined using MATLAB. The purpose of polynomial fitting was to obtain smooth cumulative heat functions for use in thermal simulation and numerical temperature prediction. The polynomial fitting methodology is depicted in Figure 3.16.

The quality of the polynomial fit was evaluated using:

R^2 value

RMSE

Agreement with measured final cumulative heat

Smoothness of the fitted curve

Physical consistency of increasing cumulative heat trend

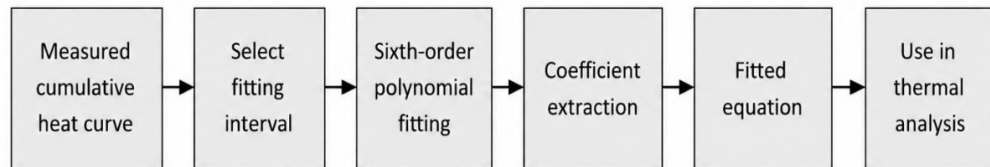


Figure 3.16: Polynomial fitting methodology chart

3.11 VCCTL Simulation

3.11.1 Purpose of VCCTL Simulation

The NIST Virtual Cement and Concrete Testing Laboratory was used to validate the experimental hydration results. VCCTL is a microstructure-based computational platform that simulates cement hydration, heat evolution, and microstructure development using cement phase composition and mixture parameters.

The purpose of using VCCTL was to:

Independently predict hydration behavior.

Validate experimental cumulative heat trends.

Compare simulated heat evolution with measured heat evolution.

Assess whether XRD-derived cement phase information can support hydration modeling.

Evaluate the applicability of VCCTL for Portland and composite cements used in Bangladesh.

3.11.2 VCCTL Input Parameters

The VCCTL simulation required cement composition and mixture parameters. XRD-derived phase information was used to prepare the cement phase input. Before entry into VCCTL, clearly inert phases, such as quartz and part of the calcium carbonate fraction, were treated separately where required, because VCCTL primarily uses clinker-based phase information for hydration simulation.

Table 3.3: VCCTL input parameters

Input parameter	Description
Cement type	CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, CEM II B-M 42.5 N
Clinker phase composition	Derived from XRD phase analysis
Major phases	C ₃ S, C ₂ S, C ₃ A, C ₄ AF or equivalent phases
Inert / filler phases	CaCO ₃ , SiO ₂ , and other minor phases where applicable
Water-to-cement ratio	≈ 0.406
Curing temperature	20°C and 25°C
Simulation duration	Up to 168 hours
Output parameters	Degree of hydration, heat rate, cumulative heat

3.11.3 VCCTL Output and Comparison

The main outputs obtained from VCCTL were:

Degree of hydration versus time

Heat generation rate versus time

Cumulative heat release versus time

These outputs were compared with the experimental results obtained from semi-adiabatic calorimetry. The comparison was conducted at selected hydration times, including 6 h, 12 h, 24 h, 48 h, 72 h, and 168 h. The VCCTL input parameters used in this study are listed in Table 3.3, and the simulation workflow is presented in Figure 3.17.

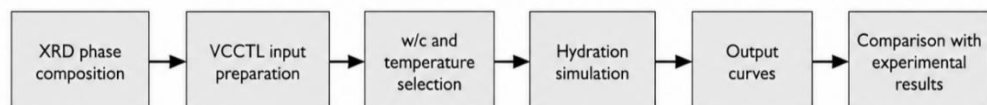


Figure 3.17: VCCTL simulation workflow chart

3.12 Validation Criteria

The experimental and simulated results were compared using both qualitative and quantitative criteria. The following parameters were compared:

Peak heat generation rate

Time of peak heat generation
 Final cumulative heat release
 Cumulative heat at selected ages
 General shape of the heat generation curve
 Degree of hydration trend
 Ranking of cement types based on heat evolution

The mean absolute error was calculated using Equation 3.25:

$$MAE = (1/n)\sum|Q_{measured} - Q_{predicted}| \quad 3.25$$

The percentage deviation was calculated using Equation 3.26:

Percentage deviation:

$$PD = \left[\frac{Q_{predicted} - Q_{measured}}{Q_{measured}} \right] \times 100 \quad 3.26$$

The validation was considered acceptable when the simulated curve followed the general shape, peak timing, and cumulative heat trend of the experimental curve.

3.13 Quality Control Measures

The following quality control measures were adopted to improve reliability:

Several quality control measures were adopted to ensure consistency and reliability throughout the experimental and analytical work. After collection, all cement samples were sealed and stored carefully to reduce moisture exposure, prehydration, and carbonation. The same mixing procedure was followed for all paste specimens to maintain uniformity among the tests. Before hydration testing, the calorimeter was calibrated, and both sample temperature and ambient temperature were recorded simultaneously during the experiment. Heat-loss correction was applied using the calibrated heat-loss coefficient, and the recorded temperature data were smoothed before numerical differentiation to reduce noise-related errors.

The same MATLAB processing procedure was applied to all cement types to ensure fair comparison of heat generation rate and cumulative heat release. During Gaussian fitting, the fitting parameters were constrained within reasonable physical limits to avoid unrealistic peak representation. Polynomial fitting was evaluated using goodness-of-fit indicators such as (R^2) and RMSE. For numerical validation, VCCTL

inputs were prepared consistently from XRD-derived phase information, and the experimental and simulated hydration curves were compared using the same selected time points. These measures helped improve the accuracy, repeatability, and reliability of the hydration heat analysis.

3.14 Limitations of Methodology

The methodology has some limitations that should be acknowledged. First, the XRD phase estimation was semi-quantitative and based on scale factor normalization. Therefore, the phase percentages should be treated as approximate values rather than exact phase compositions.

Second, the semi-adiabatic calorimeter was custom-built. As a result, the accuracy of the calculated heat evolution depends strongly on the accuracy of calibration parameters such as the effective heat capacity and the heat-loss coefficient.

Third, VCCTL requires idealized material inputs, including phase composition, fineness, and particle size distribution. Some of these values were derived from XRD data, while others may be taken from supplier information or reasonable assumptions where direct measurement was unavailable.

Fourth, Gaussian and polynomial models are mathematical fitting tools. They are useful for representing measured heat evolution, but they do not independently describe all chemical reactions occurring during cement hydration.

Despite these limitations, the combined use of XRD, semi-adiabatic calorimetry, MATLAB modeling, and VCCTL validation provides a comprehensive methodology for evaluating the hydration behavior of Portland and composite cements.

3.15 Summary

This chapter described the methodology adopted to investigate the hydration behavior of Portland and composite cements under semi-adiabatic conditions. Four commercially available cement types conforming to BDS EN 197-1:2003 were selected, namely CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N. XRD analysis was used to identify major mineral phases, and semi-adiabatic calorimetry was used to monitor hydration temperature development under sealed conditions.

The measured temperature data were processed using MATLAB to calculate

heat generation rate and cumulative heat release. A two-component Gaussian model was used to represent the major hydration peaks, while sixth-order polynomial fitting was used to obtain continuous cumulative heat equations. The De Schutter–Taerwe model was implemented to predict hydration, and the VCCTL simulation was used to validate experimental trends.

The methodology developed in this chapter forms the basis for the analysis and discussion of the evolution of hydration heat, model fitting, and simulation validation in the next chapter.

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents the experimental and computational results obtained from the investigation of hydration behavior of Portland and composite cements under semi-adiabatic conditions. The results are discussed based on X-ray diffraction (XRD) characterization, semi-adiabatic calorimetric measurements, heat generation rate, cumulative heat release, Gaussian peak fitting, polynomial modeling, and VCCTL-based simulation.

The main purpose of this chapter is to compare the hydration behavior of CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N. The comparison focuses on identifying differences in heat evolution, hydration kinetics, peak heat generation rate, cumulative heat release, and suitability for mass concrete applications. Since excessive hydration heat can increase the risk of thermal cracking in mass concrete, the thermal performance of each cement type is evaluated carefully.

The results show that CEM I 52.5 N produced the highest heat release and fastest hydration response, while CEM II B-M 42.5 N showed the lowest heat release and slowest hydration behavior. CEM II A-M 42.5 N exhibited intermediate behavior between the Portland cement and the higher blended composite cement. CEM II A-M 42.5 R caused a substantially higher early-age heat generation rate than would normally be expected from a composite cement. The XRD results are presented in Section 4.2, calorimetric results in Section 4.3, VCCTL simulation results in Section 4.4, and the combined discussion in Section 4.5.

4.2 XRD Analysis and Mineralogical Characterization

X-ray diffraction analysis was performed to identify the major crystalline phases present in the cement samples. The diffractograms of the four samples are shown in Figures 4.1 to 4.4. The identified phases are important because the heat of cement hydration is strongly influenced by clinker mineralogy. Alite, or C_3S , is primarily responsible for early hydration and early strength development, while belite, or C_2S , contributes more slowly to long-term hydration. Aluminate phases hydrate rapidly and

are responsible for early exothermic reactions. Calcium carbonate and quartz primarily act as filler phases, reducing the effective clinker content in composite cements.

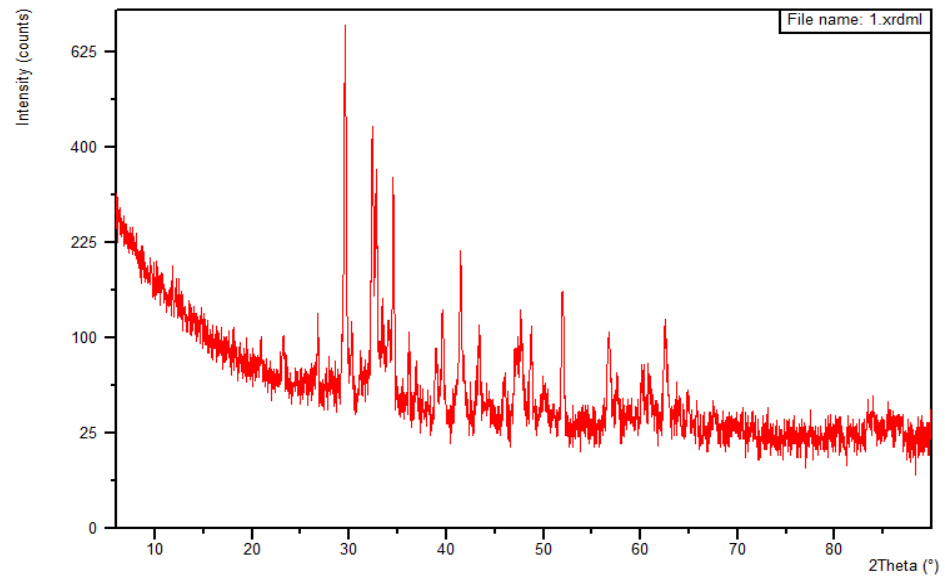


Figure 4.1: XRD diffraction pattern of Cement 1 (CEM II B-M 42.5 N)

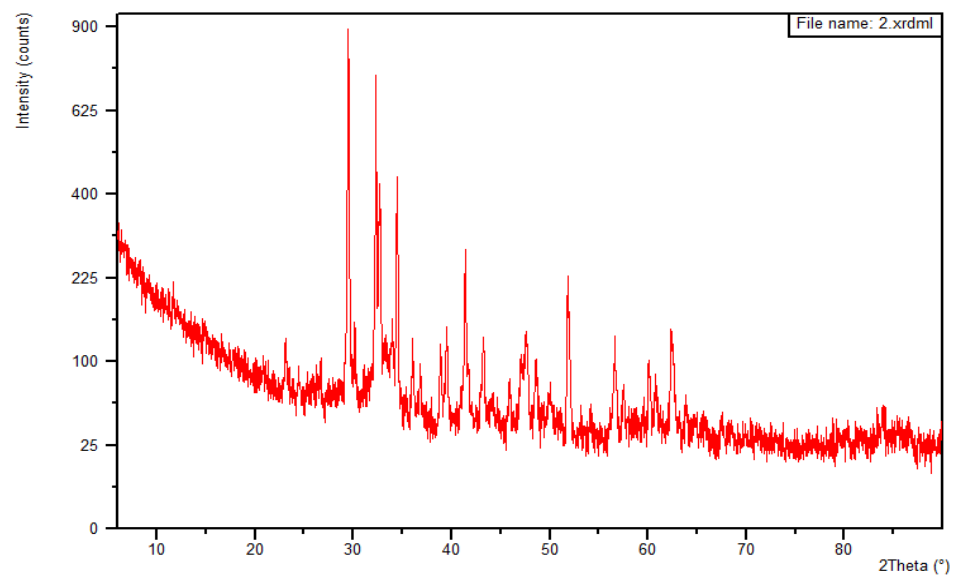


Figure 4.2: XRD diffraction pattern of Cement 2 (CEM II A-M 42.5 N)

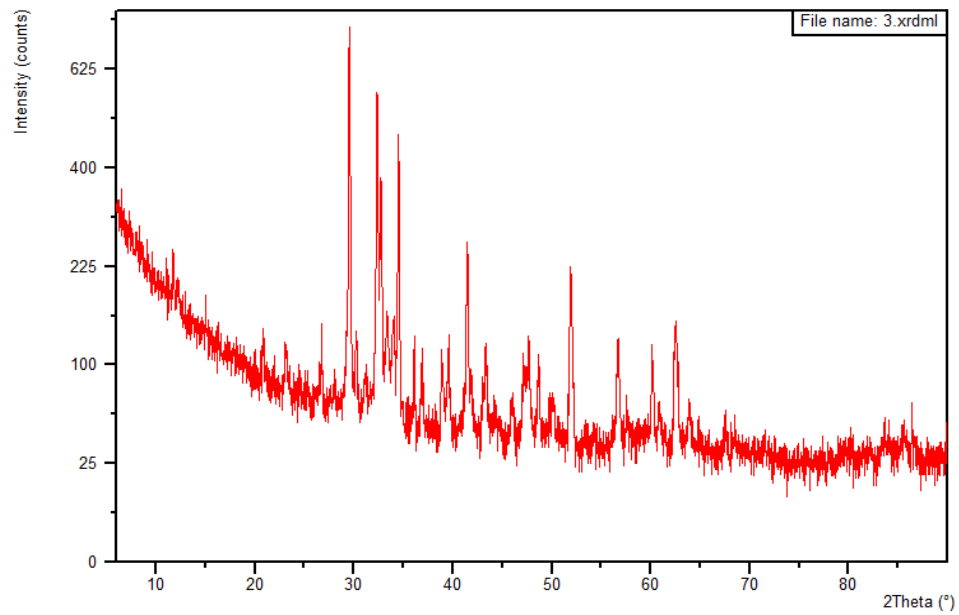


Figure 4.3: XRD diffraction pattern of Cement 3 (CEM II A-M 42.5 R)

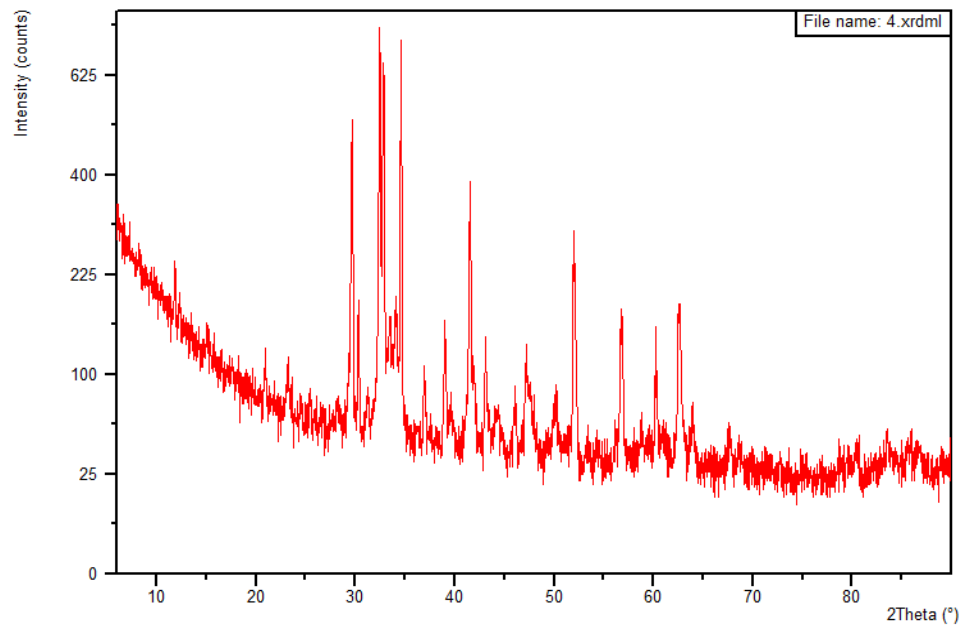


Figure 4.4: XRD diffraction pattern of Cement 4 (CEM I 52.5 N)

The XRD patterns showed the presence of calcium silicate phases, calcium carbonate, quartz, aluminate phases, and calcium-magnesium-aluminum silicate phases. Calcium carbonate was more prominent in the composite cement samples, indicating mineral additions. The semi-quantitative phase compositions of all samples are summarised in Table 4.1, and the corresponding phase distribution is shown in Figure 4.5. This is consistent with the expected composition of CEM II-type cement.

Table 4.1: Semi-quantitative XRD phase composition of cement Samples.

Phase / Compound	CEM I 52.5 N (%)	CEM II A-M 42.5 R (%)	CEM II A-M 42.5 N (%)	CEM II B-M 42.5 N (%)	Hydration relevance
C ₃ S (Alite)	58.8	51.5	50.7	42.4	Main contributor to early strength
C ₂ S (Belite)	17.6	14.6	14.9	12.1	Strength development at later ages
C ₃ A (Aluminate)	9.8	7.8	7.0	6.1	Very reactive; affects setting and early hydration
C ₄ AF (Ferrite)	9.8	7.8	7.0	6.1	Minor phase; influences hydration and color
Gypsum (Hemihydrate)	3.9	3.9	3.5	3.0	Controls setting time by regulating aluminate reactivity
Filler/SCM	—	14.4	16.9	30.3	Silicate/mineral addition

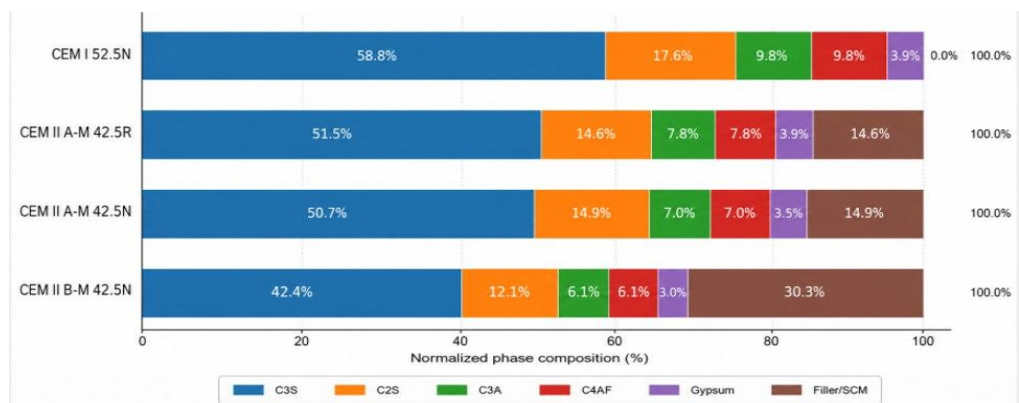


Figure 4.5: Bar chart showing semi-quantitative XRD phase composition

The XRD results indicate that the hydration behavior of the selected cements is controlled by differences in clinker content and mineral additions. Samples containing higher calcium silicate phases are expected to show stronger hydration heat evolution, while samples with higher carbonate or supplementary phases are expected to generate lower heat due to clinker dilution. Therefore, the XRD results provide a mineralogical explanation for the differences observed in the calorimetric results.

4.3 Semi-Adiabatic Calorimetric Measurements

Heat evolution during cement hydration was monitored using a semi-adiabatic calorimeter with a calibrated insulator ($C_{\text{insulator}} = 1386.6 \text{ J/K}$; combined system capacity = 3076.9 J/K). Cement pastes were prepared at a water-to-cement ratio of 0.40. The raw thermocouple data were processed in MATLAB using: (i) a two-component Gaussian model to resolve the multi-stage heat rate, and (ii) a sixth-order polynomial fit to the cumulative heat curve for use in numerical thermal simulations. Results for each cement type are presented individually below, followed by a comparative analysis.

The processed heat-generation-rate and cumulative-heat curves for all four cements are presented together From Figure 4.6 and Figure 4.8 so that the samples can be compared directly. In each figure the experimentally derived curve (left panel) is shown alongside the corresponding two-component Gaussian (Gauss-2) reconstruction (right panel). The individual per-sample heat-rate, Gaussian-decomposition and Polynomial-6 plots are provided in Appendix D and Appendix E, while the fitted Gaussian parameters and terminal cumulative-heat values for all samples are summarized in Table 4.2.

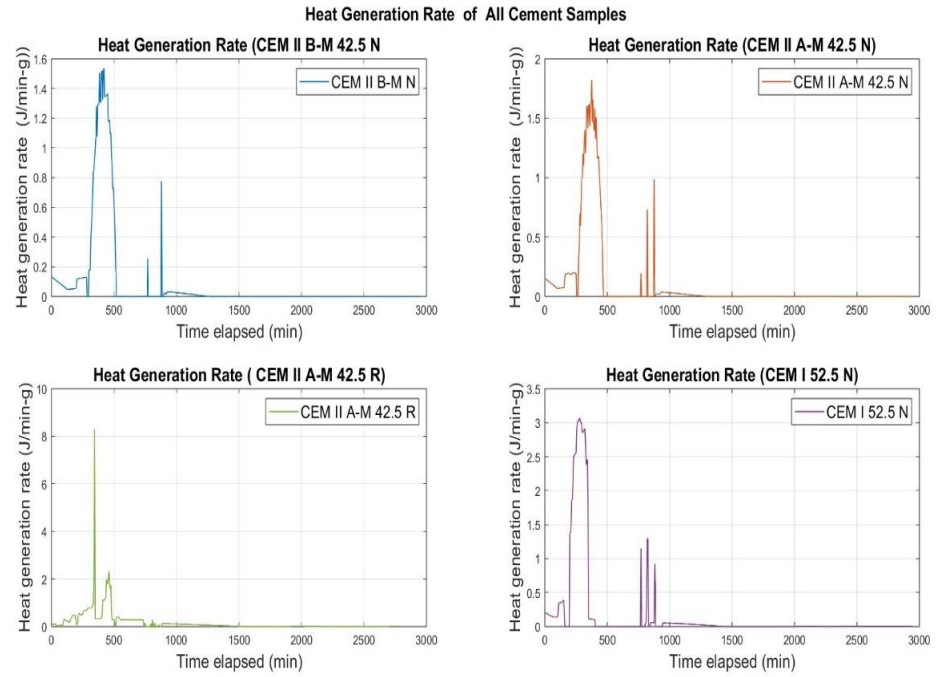


Figure 4.6: MATLAB heat generation rate for all cement samples
(Experimental)

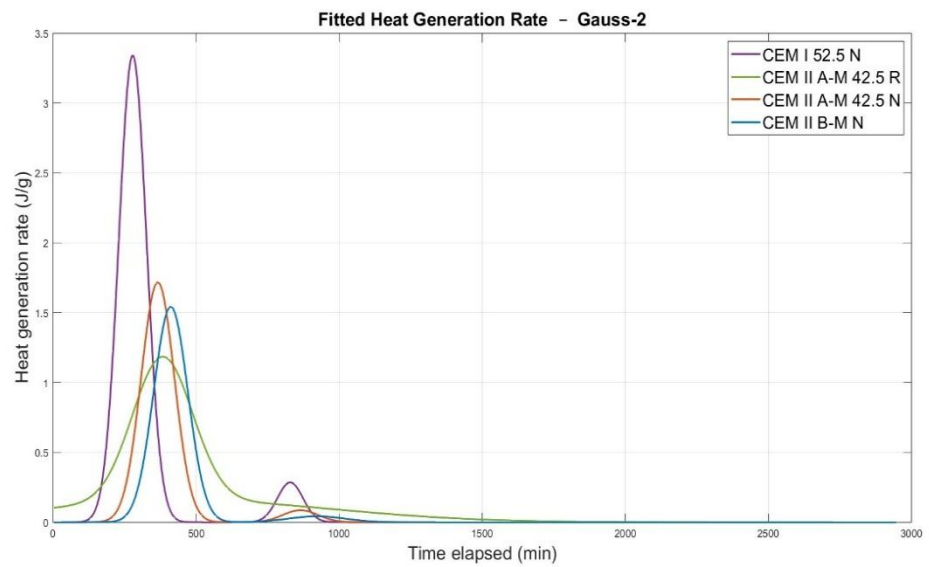


Figure 4.7: MATLAB heat generation rate for all cement samples
(Fitted Data – Gauss 2)

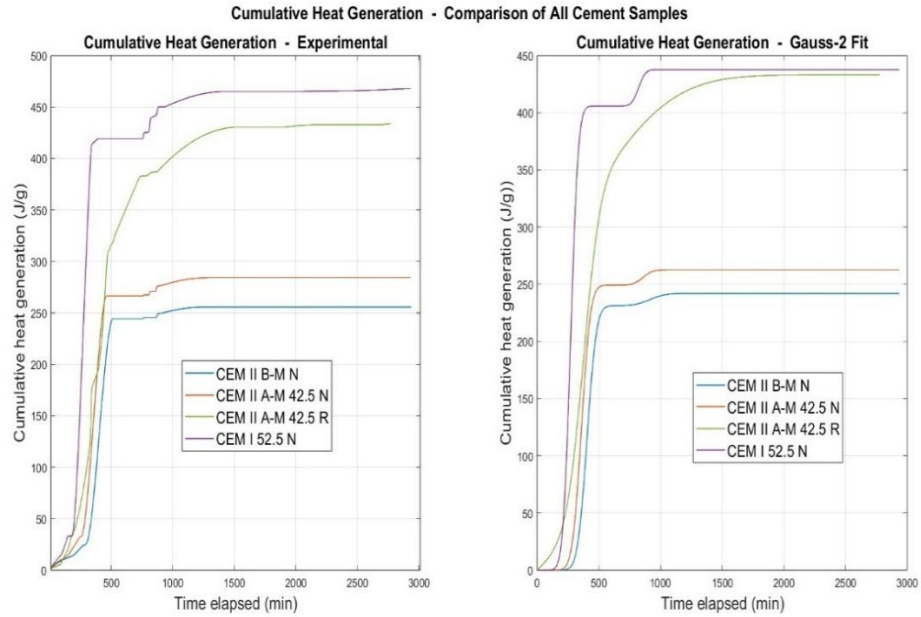


Figure 4.8 Combined MATLAB cumulative heat generation for all cement samples experimental (left) and two-component Gaussian (Gauss-2) fit (right).

CEM I 52.5 N exhibited the fastest hydration kinetics and the most intense main silicate peak among the tested cements, with a fitted G_2 amplitude of approximately $3.3 \text{ J} \cdot \text{min}^{-1} \cdot \text{g}^{-1}$ near 290 min and a terminal cumulative heat of about 467 J/g (experimental) and 438 J/g (Gauss-2 fit). This behavior is consistent with its high C_3S content and the absence of clinker dilution (Table 4.2).

CEM II A-M 42.5 R showed the second-highest early-age reactivity. Although its mineral-addition content is comparable to that of CEM II A-M 42.5 N, its rapid-hardening (“R”) classification achieved primarily through increased fineness produced a broad, high silicate peak and a terminal cumulative heat of about 433 J/g (Appendix D), placing its thermal behavior closer to CEM I 52.5 N than to the normal-strength composite cement.

CEM II A-M 42.5 N displayed intermediate behavior, with the main silicate peak retarded relative to CEM I owing to clinker dilution by limestone and other supplementary constituents (6–20% replacement). Its terminal cumulative heat reached approximately 284 J/g (experimental) and 262 J/g (Gauss-2 fit).

CEM II B-M 42.5 N showed the slowest hydration kinetics, consistent with its higher replacement level (21–35%). Its G_2 silicate peak occurred latest, and its cumulative heat plateaued at approximately 255 J/g (experimental) and 242 J/g (Gauss-2 fit) the lowest among the tested cements making it the most suitable choice for thermally sensitive mass-concrete applications.

Table 4.2: Gaussian Model Parameters and Cumulative Heat Release for All Cement Types

Parameter	CEM I 52.5N	CEM II A-M 42.5N	CEM II B-M 42.5N	CEM II A-M 42.5R	Unit
G ₁ Amplitude (A ₁)	~0.40–0.45	~0.15–0.20	~0.12–0.16	~0.30–0.40	J/min-g
G ₁ Peak Time (μ_1)	~220–260	~280–320	~320–360	~230–270	min
G ₂ Amplitude (A ₂)	~2.8–3.3	~1.5–2.0	~1.3–1.6	~2.2–2.6	J/min-g
G ₂ Peak Time (μ_2)	~280–340	~400–460	~450–510	~310–370	min
Cumuli. heat (terminal)	~440–480	~260–285	~245–265	~430–460	J/g
Fit order	Gaussian 2	Gaussian 2	Gaussian 2	Gaussian 2	

4.4 VCCTL Simulation and Validation

The NIST Virtual Cement and Concrete Testing Laboratory (VCCTL) was used to independently simulate the degree of hydration (DOH) and heat release for each cement type. Clinker phase proportions derived from XRD analysis were used as primary input parameters, along with particle size distribution and a water-to-cement ratio of 0.40. The simulated DOH and cumulative heat release curves are compared against the experimentally derived De Schutter–Taerwe hydration model in the following subsections.

The VCCTL-simulated degree of hydration (DOH) and cumulative heat release for all four cements are compared in Figure 4.9 and Figure 4.10; the corresponding per-sample simulation plots are provided in Appendix E. All simulations were run under isothermal conditions at a water-to-cement ratio of 0.40, using clinker-phase proportions derived from the XRD analysis (Table 4.1) as the primary inputs. The simulated trends are summarized in Table 4.3 and are compared below against the experimentally derived behavior of Section 4.3.

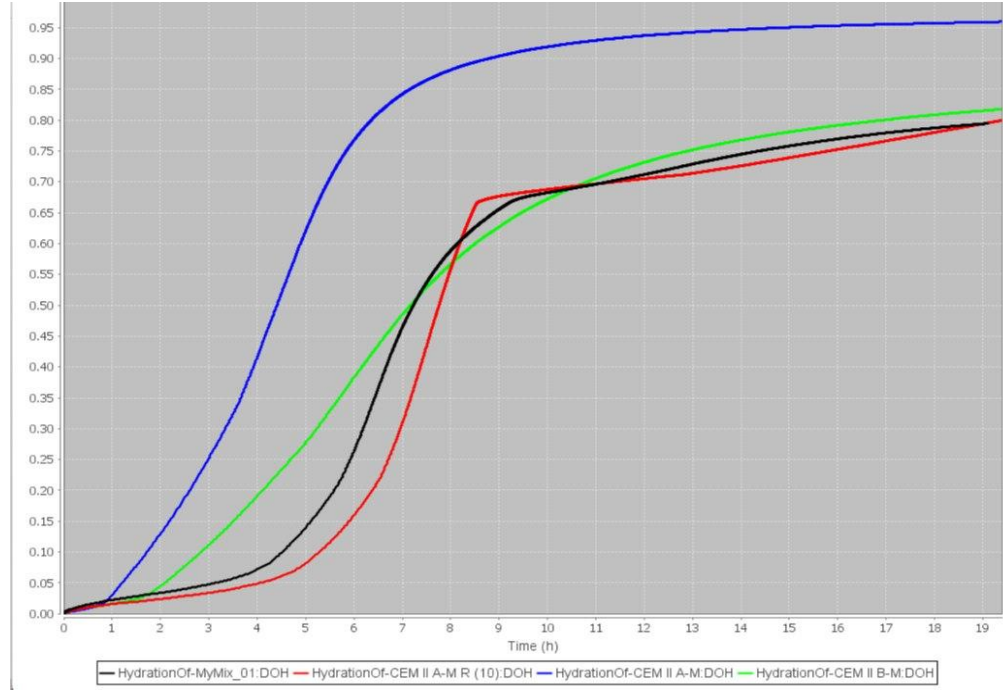


Figure 4.9: Combined VCCTL-simulated degree of hydration versus time for all four cements (isothermal, $w/c = 0.40$).

Legend mapping: MyMix_01 = CEM I 52.5 N; CEM II A-M = CEM II A-M 42.5 N; CEM II A-M R (10) = CEM II A-M 42.5 R; CEM II B-M = CEM II B-M 42.5 N.

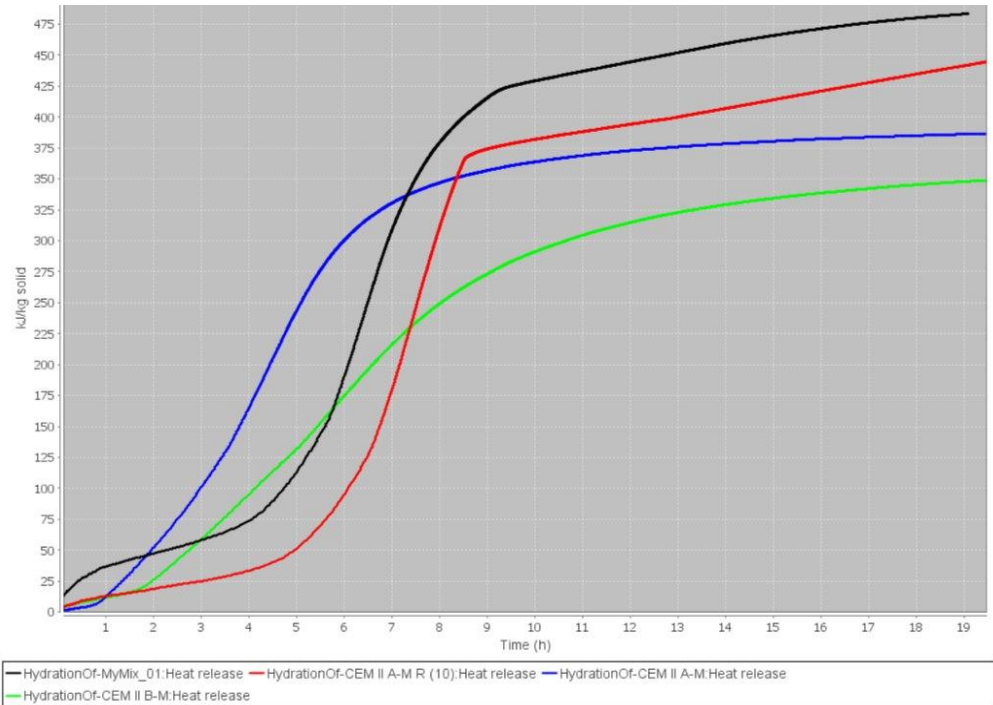


Figure 4.10: Combined VCCTL-simulated cumulative heat release versus time for all four cements (isothermal, $w/c = 0.40$).

Legend mapping: MyMix_01 = CEM I 52.5 N; CEM II A-M = CEM II A-M 42.5 N; CEM II

$$A-M R (10) = CEM II A-M 42.5 R; CEM II B-M = CEM II B-M 42.5 N.$$

CEM II A-M 42.5 N reached the highest simulated degree of hydration (≈ 0.95 at 19 h), with the earliest acceleration (3–5 h). This value is higher than that inferred from the semi-adiabatic experiment because VCCTL operates under isothermal conditions, which typically yield greater ultimate hydration than heat-loss-corrected semi-adiabatic tests.

CEM I 52.5 N showed a characteristic sigmoidal DOH profile reaching ≈ 0.81 at 19 h, with rapid acceleration between 5–10 h corresponding to the main silicate stage, and the highest simulated heat release of ≈ 480 kJ/kg. CEM II A-M 42.5 R reached a DOH of ≈ 0.80 with the second-highest simulated heat release (≈ 445 kJ/kg), consistent with its elevated early-age reactivity. CEM II B-M 42.5 N reached a DOH of ≈ 0.825 yet the lowest heat release (≈ 348 kJ/kg), reflecting its reduced clinker content and the retarding effect of supplementary constituents.

Overall, the VCCTL predictions reproduce the experimental ranking of thermal output and confirm the mineralogical interpretation of Section 4.2. The minor differences in absolute magnitude are attributable to the semi-adiabatic idealization of the simulation and to the semi-quantitative nature of the XRD-derived phase inputs (Table 4.3).

Table 4.3: VCCTL Simulation Summary: Degree of Hydration and Heat Release at 19h

Cement / Mix	VCCTL DOH at 19 h (–)	VCCTL Heat Release at 19 h (kJ/kg)	Condition
CEM I 52.5N	≈ 0.81	≈ 480	Isothermal
CEM II A-M 42.5N	≈ 0.950	≈ 390	Isothermal
CEM II B-M 42.5N	≈ 0.825	≈ 348	Isothermal
CEM II A-M 42.5R	≈ 0.800	≈ 445	Isothermal

4.5 Discussion

4.5.1 Clinker Composition — Hydration Kinetics Correlation

Table 4.4: Hydration Kinetics Correlation

Cement Type	Relative Heat Generation Rate	Explanation
CEM I 52.5 N	Highest	Contains the highest clinker content and exhibits the fastest hydration kinetics, producing the largest silicate hydration peak and the highest cumulative heat release (~467 J/g).
CEM II A-M 42.5 R	High	Although it contains mineral additions similar to those of CEM II A-M 42.5 N, its rapid-hardening ("R") classification is achieved primarily through increased fineness and proprietary performance-enhancing measures. The larger specific surface area accelerates clinker hydration, resulting in a substantially higher early-age heat generation rate than would normally be expected for a composite cement.
CEM II A-M 42.5 N	Moderate	Lower clinker content than CEM I and normal-strength development. Mineral additions dilute clinker content and slow hydration, resulting in an intermediate heat evolution profile.
CEM II B-M 42.5 N	Lowest	Highest proportion of mineral constituents and lowest clinker content among the four cements. Consequently, it exhibits the slowest hydration kinetics and the lowest cumulative heat release (~255 J/g).

4.5.2 Thermal Cracking Risk Assessment

From a practical standpoint, cumulative heat data directly determines the risk of thermal cracking in mass concrete. ACI 207R and CIRIA C660 recommend limiting the temperature differential between the core and surface of a mass concrete element to approximately 20°C to minimize the risk of early-age thermal cracking. The correlation

between clinker composition and hydration kinetics for each cement is summarized in Table 4.4.

Based on the measured and modeled cumulative heat release, the estimated adiabatic temperature rise ($\Delta T_{ad_{ia\beta}} \approx Q/C_{p,paste}$, where $C_{p,paste} \approx 1.0 \text{ kJ/kg} \cdot \text{K}$) varies significantly among the investigated cements. CEM I 52.5 N exhibited the highest cumulative heat release (approximately 467 J/g), corresponding to an estimated adiabatic temperature rise exceeding 40–45°C and indicating a high risk of thermal cracking in thick concrete sections unless adequate temperature-control measures are implemented.

Although CEM II A-M 42.5 R contains mineral additions similar to CEM II A-M 42.5 N, its rapid-hardening classification is achieved through increased fineness and enhanced hydration characteristics, resulting in substantially greater early-age heat evolution. Consequently, the estimated adiabatic temperature rise for CEM II A-M 42.5 R is expected to be approximately 32–38°C, indicating a moderate-to-high thermal cracking risk. While lower than that of CEM I 52.5 N, it remains significantly higher than the corresponding value for normal-strength composite cement.

For CEM II A-M 42.5 N, the reduced clinker content and slower hydration kinetics lower the estimated adiabatic temperature rise to approximately 25–28°C, thereby reducing the likelihood of excessive thermal gradients. The lowest temperature rise was observed for CEM II B-M 42.5 N, with an estimated range of 22–25°C due to its higher proportion of supplementary cementitious constituents and slower hydration rate.

Overall, the thermal cracking risk may be ranked as:

CEM I 52.5 N > CEM II A-M 42.5 R > CEM II A-M 42.5 N > CEM II B-M 42.5 N

indicating that composite cements, particularly CEM II B-M 42.5 N, are more suitable for mass concrete applications where temperature control is critical. Nevertheless, the use of CEM II A-M 42.5 R in large concrete pours should be accompanied by appropriate thermal management measures, as its accelerated hydration can generate considerably higher early-age temperatures than those of conventional CEM II A-M 42.5 N. These considerations are especially important in the tropical climatic conditions of Bangladesh, where elevated ambient temperatures may further increase internal thermal gradients and the potential for thermal cracking.

4.5.3 Polynomial Model Applicability

The sixth-order polynomial functions (Equation 3.23) fitted to the cumulative heat curves achieve excellent goodness-of-fit ($R^2 > 0.999$) and provide continuous, differentiable analytical expressions that can be directly incorporated into finite element thermal simulations of early-age concrete structures. Unlike the Gaussian rate model, the polynomial cumulative heat function is well-conditioned over the full monitoring period and avoids extrapolation instability beyond the fit window. Both representations are complementary: the Gaussian model is preferred for mechanistic interpretation, while the polynomial fit is more practical for numerical simulation.

4.5.4 Limitations and Future Work

The semi-adiabatic method introduces a heat-loss correction calibrated using an independent cooling experiment. While the calibrated insulator heat capacity ($C_{\text{insulator}} = 1386.6 \text{ J/K}$) reduces uncertainty to approximately $\pm 3\%$, this correction is sensitive to ambient temperature, so experiments conducted at different room temperatures should recalibrate the constant. The calibration procedure is based on the lumped thermal model was fitted to the cooling data, as expressed by Equation 4.1.

$$\frac{dq}{dt} = \frac{1}{m_c} [C_{cys} \frac{dT}{dt} + K_l (T - T_{amb})] \quad 4.1$$

Future work should extend measurements to additional water-to-cement ratios (0.30, 0.45, 0.50), apply full Rietveld XRD refinement for more accurate phase quantification, and incorporate a temperature-dependent Arrhenius correction to the hydration rate to enable field-condition predictions across ambient temperatures relevant to Bangladesh.

4.6 Chapter Summary

This chapter presented the complete experimental results and analysis for four BDS EN 197-1:2003-compliant cement types: CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N. The principal findings are summarized as follows:

XRD analysis identified C_3S as the dominant mineralogical phase in all samples (Table 4.1; Figures 4.1–4.4). CEM I 52.5N shows the highest alite content while CEM II B-M exhibits increased proportions of $CaCO_3$ and silicate phases consistent with higher SCM replacement, as illustrated in the phase composition bar chart (Figure 4.5).

Under normal circumstances, one might expect a composite cement such as CEM II A-M 42.5 R to generate significantly less heat than CEM I 52.5 N because of its lower clinker content. However, Early strength has been gained by increasing cement fineness (higher Blaine surface area), optimizing clinker mineralogy (higher reactive C_3S content), using grinding aids or hydration-enhancing additives, and improving particle-size distribution.

MATLAB two-component Gaussian decomposition (Figure 4.6 to 4.8); Table 4.2) successfully resolved the overlapping C_3A (G_1) and C_3S (G_2) hydration peaks. CEM I 52.5N exhibits the earliest and most intense G_2 peak; CEM II B-M 42.5N shows a systematically retarded and attenuated G_2 peak, directly linked to its lower clinker fraction.

VCCTL simulations under isothermal conditions (Figures 4.9–4.10; Table 4.3) confirmed the trends in hydration behavior identified by the semi-adiabatic experiments. Predicted DOH and heat release values are consistent with XRD-derived clinker phase proportions.

CEM I 52.5N poses the highest thermal cracking risk (estimated adiabatic $\Delta T > 40^\circ\text{C}$), while CEM II B-M 42.5N offers the lowest risk ($\sim 22\text{--}25^\circ\text{C}$), making composite cements significantly more suitable for mass concrete applications in tropical environments.

The calibrated semi-adiabatic apparatus, MATLAB processing pipeline, and VCCTL validation framework together constitute a robust experimental methodology for cement hydration characterization that can be applied to future cement types and blend designs without modification.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1 General

This study investigated the hydration behavior of four commercially available cements conforming to BDS EN 197-1:2003 — CEM I 52.5 N, CEM II A-M 42.5 R, CEM II A-M 42.5 N, and CEM II B-M 42.5 N — under semi-adiabatic and sealed conditions. A combined experimental, analytical, and numerical framework integrating XRD characterization, semi-adiabatic calorimetry, MATLAB-based Gaussian and polynomial modeling, and VCCTL simulation was applied to quantify hydration heat evolution and assess thermal cracking risk for each cement type.

The results confirmed a clear heat-evolution ranking:

CEM I 52.5 N (≈ 467 J/g) > CEM II A-M 42.5 R (≈ 433 J/g) > CEM II A-M 42.5 N (≈ 284 J/g) > CEM II B-M 42.5 N (≈ 255 J/g). Notably, the rapid-hardening composite CEM II A-M 42.5 R released heat approaching CEM I level, a consequence of its higher Blaine fineness rather than clinker content — a finding with direct practical implications for mass concrete placement.

A quantitative comparison between experimental terminal cumulative heat and VCCTL-simulated heat release at 19 h is summarized in Table 5.1. VCCTL agrees closely with experiment for high-clinker cements (relative difference $\approx +2.8\%$ for CEM I 52.5 N and $+2.7\%$ for CEM II A-M 42.5 R), while over-predicting slow blended cements by approximately 37% (mean absolute deviation $\approx 20\%$). The systematic over-prediction for blended cements is physically expected: VCCTL operates under isothermal conditions approaching ultimate degree of hydration, whereas the semi-adiabatic test is heat-loss-corrected and time-truncated. Residual scatter is consistent with the $\pm 3\%$ calorimeter calibration uncertainty ($C_v = 1386.6$ J/K; combined system capacity = 3076.9 J/K).

Table 5.1: Comparison of experimental and VCCTL cumulative heat with relative difference.

Cement	Exp. Q (J/g)	VCCTL Q at 19 h (kJ/kg)	Rel. diff. (%)	Primary cause of deviation
CEM I 52.5 N	≈ 467	≈ 480	+2.8	Semi-adiabatic captures the full early silicate peak; VCCTL output truncated at 19 h.
CEM II A-M 42.5 N	≈ 284	≈ 390	+37.3	Isothermal VCCTL reaches DOH ≈ 0.95; slower sealed reaction limits measured heat.
CEM II A-M 42.5 R	≈ 433	≈ 445	+2.7	High fineness; isothermal ultimate heat slightly exceeds the truncated experimental value.
CEM II B-M 42.5 N	≈ 255	≈ 348	+36.5	Low clinker content; isothermal conditions accelerate the otherwise slow blended hydration.

5.2 Limitations of the Study

The following limitations should be acknowledged when interpreting the results:

XRD phase quantification used the semi-quantitative scale-factor method; a full Rietveld refinement would yield more accurate clinker percentages and improve VCCTL simulation inputs.

Testing was conducted on cement paste only. Aggregate type, admixtures, member dimensions, and field curing conditions all influence temperature development in real concrete and were not captured at paste scale.

The custom semi-adiabatic calorimeter has an estimated calibration uncertainty of ±3%; any drift in the effective heat capacity (C) or heat-loss coefficient (K_i) propagates directly into the calculated heat values.

Only four commercial cement samples from specific production batches were tested; results may not generalize to all brands or batch variations available in Bangladesh.

Gaussian and polynomial models are mathematical fitting tools; they represent measured curves with high fidelity but do not independently resolve all underlying chemical reaction mechanisms.

5.3 Conclusions

The three principal conclusions of this study are as follows:

Cumulative heat ranking and thermal cracking risk: The experimental terminal heat release ranks as CEM I 52.5 N (≈ 467 J/g, $\Delta T > 40$ °C) > CEM II A-M 42.5 R (≈ 433 J/g, $\Delta T \approx 32$ – 38 °C) > CEM II A-M 42.5 N (≈ 284 J/g, $\Delta T \approx 25$ – 28 °C) > CEM II B-M 42.5 N (≈ 255 J/g, $\Delta T \approx 22$ – 25 °C). CEM II B-M 42.5 N, releasing $\sim 45\%$ less heat than CEM I, is the most thermally favorable option for mass concrete in Bangladesh.

Rapid-hardening composite cement behaves near Portland levels: Despite containing mineral additions comparable to CEM II A-M 42.5 N, the R-class cement released ≈ 433 J/g — only 7% below CEM I and 52% above CEM II A-M 42.5 N — because increased Blaine fineness accelerates C_3S hydration kinetics. Engineers must not assume that a CEM II A-M 42.5 R cement is thermally equivalent to a normal-hardening CEM II composite; it requires the same thermal management precautions as ordinary Portland cement in thick placements.

VCCTL validation confirms experimental ranking: VCCTL agrees with experiment within ≈ 2.7 – 2.8% for high-clinker cements (CEM I 52.5 N and CEM II A-M 42.5 R) but over-predicts by $\approx 37\%$ for slow blended cements (CEM II A-M 42.5 N and CEM II B-M 42.5 N), consistent with isothermal simulation conditions and a mean absolute deviation of $\approx 20\%$. The experimental ranking is fully reproduced, validating the XRD–calorimetry–MATLAB–VCCTL framework as a reliable methodology for cement hydration characterization.

5.4 Practical Implications

The measured heat-evolution data provide direct guidance for cement selection in temperature-sensitive construction. CEM I 52.5 N is suitable where high early strength is required, but mandatory temperature-control measures (pre-cooling, staged casting, insulation management) must be adopted for sections exceeding 0.5 m in thickness. CEM II A-M 42.5 R offers moderately faster strength gain than CEM II A-M 42.5 N but carries near-Portland thermal risk; it should be treated as a high-heat cement in mass concrete design. CEM II A-M 42.5 N is appropriate for general structural concrete where a balance between strength development and heat reduction is needed. CEM II B-M 42.5 N, with the lowest cumulative heat (≈ 255 J/g) and slowest hydration, is the preferred choice for raft foundations, bridge piers, and other mass concrete elements in the tropical climate of Bangladesh, where elevated ambient temperatures further amplify thermal gradients.

5.5 Recommendations for Future Work

Future research should address the following priorities:

Extended material and condition testing: evaluate additional cement brands, water-to-cement ratios (0.30, 0.45, 0.50), and curing temperatures representative of Bangladeshi field conditions to develop a broader hydration heat database.

Advanced characterization: apply full Rietveld XRD refinement and X-ray fluorescence for more accurate phase quantification, supplemented by SEM, TGA, and FTIR to characterize hydration products and microstructural development.

Scale-up validation: conduct calorimetric tests on mortar and concrete specimens and validate the developed Gaussian and polynomial heat models against temperature-monitoring data from real mass concrete placements (raft slabs, bridge foundations).

Thermal modelling and FEM integration: embed the polynomial cumulative heat functions directly into finite element thermal simulation software and apply Arrhenius-based temperature corrections for field condition predictions.

Practical guidelines: develop a structured cement selection framework for mass concrete construction in Bangladesh, combining hydration heat data, structural geometry, ambient temperature, and acceptable thermal cracking risk into a decision

tool for practicing engineers.

REFERENCES

- Bentz, D. P. (2007). Transient plane source measurements of the thermal properties of hydrating cement pastes. *Materials and Structures*, 40, 1073–1080. <https://doi.org/10.1617/s11527-006-9206-9>
- Bullard, J. W., Jennings, H. M., Livingston, R. A., Nonat, A., Scherer, G. W., Schweitzer, J. S., Scrivener, K. L., & Thomas, J. J. (2011). Mechanisms of cement hydration. *Cement and Concrete Research*, 41(12), 1208–1223. <https://doi.org/10.1016/j.cemconres.2010.09.011>
- De Schutter, G., & Taerwe, L. (1995a). General hydration model for Portland cement and blast furnace slag cement. *Cement and Concrete Research*, 25(3), 593–604. [https://doi.org/10.1016/0008-8846\(95\)00048-8](https://doi.org/10.1016/0008-8846(95)00048-8)
- De Schutter, G., & Taerwe, L. (1995b). Specific heat and thermal diffusivity of hardening concrete. *Magazine of Concrete Research*, 47(172), 203–208.
- Ferraro, C. C., Watts, B., Tao, C., & Masters, F. (2018). *Advanced analysis, validation and optimization of virtual cement and concrete testing* (Final Report). University of Florida, Florida Department of Transportation Research Center.
- Han, F., Liu, R., Wang, D., & Yan, P. (2014). Characteristics of the hydration heat evolution of composite binder at different hydrating temperature. *Thermochimica Acta*, 586, 52–57. <https://doi.org/10.1016/j.tca.2014.04.019>
- Han, X., Yang, J., Feng, J., Zhou, C., & Wang, X. (2019). Research on hydration mechanism of ultrafine fly ash and cement composite. *Construction and Building Materials*, 227, Article 116697. <https://doi.org/10.1016/j.conbuildmat.2019.116697>
- Hernandez-Bautista, E., Bentz, D. P., Sandoval-Torres, S., & Cano-Barrita, P. F. de J. (2016). Numerical simulation of heat and mass transport during hydration of Portland cement mortar in semi-adiabatic and steam curing conditions. *Cement and Concrete Composites*, 69, 38–48. <https://doi.org/10.1016/j.cemconcomp.2016.03.005>
- Klemczak, B., & Batog, M. (2016). Heat of hydration of low-clinker cements. Part I. Semi-adiabatic and isothermal tests at different temperatures. *Journal of Thermal Analysis and Calorimetry*, 123(1), 1351–1360. <https://doi.org/10.1007/s10973-015-5048-4>

Liu, D., Zhang, W., Tang, Y., & Jian, Y. (2021). Prediction of hydration heat of mass concrete based on the SVR model. *IEEE Access*, 9, 58889–58900.

<https://doi.org/10.1109/ACCESS.2021.3075212>

Lothenbach, B., Scrivener, K., & Hooton, R. D. (2011). Supplementary cementitious materials. *Cement and Concrete Research*, 41(12), 1244–1256.

<https://doi.org/10.1016/j.cemconres.2010.12.001>

Meinhard, K., & Lackner, R. (2008). Multi-phase hydration model for prediction of hydration-heat release of blended cements. *Cement and Concrete Research*, 38(6), 794–802.

<https://doi.org/10.1016/j.cemconres.2008.01.008>

Mills, R. H. (1966). Factors influencing cessation of hydration in water-cured cement pastes. *Highway Research Board Special Report*, 90, 406–424.

Pane, I., & Hansen, W. (2005). Investigation of blended cement hydration by isothermal calorimetry and thermal analysis. *Cement and Concrete Research*, 35(6), 1155–1164. <https://doi.org/10.1016/j.cemconres.2004.10.027>

Pang, X., Bentz, D. P., Meyer, C., Funkhouser, G. P., & Darbe, R. (2013). A comparison study of Portland cement hydration kinetics as measured by chemical shrinkage and isothermal calorimetry. *Cement and Concrete Composites*, 39, 23–32.

<https://doi.org/10.1016/j.cemconcomp.2013.03.007>

Quennoz, A., & Scrivener, K. L. (2013). Hydration of C₃A–gypsum systems. *Cement and Concrete Research*, 44, 46–54.

<https://doi.org/10.1016/j.cemconres.2012.10.017>

Rastogi, H., Moharana, S., & Subhash, S. (2024). Prediction of hydration properties and microstructure development of cement mortar using virtual cement and concrete testing laboratory (VCCTL). *Discover Civil Engineering*, 1, Article 152.

<https://doi.org/10.1007/s44290-024-00155-y>

RILEM Technical Committee 119-TCE. (1997). Adiabatic and semi-adiabatic calorimetry to determine the temperature increase in concrete due to hydration heat of cement. *Materials and Structures*, 30(8), 451–464.

<https://doi.org/10.1007/BF02524773>

Schindler, A. K., & Folliard, K. J. (2005). Heat of hydration models for cementitious materials. *ACI Materials Journal*, 102(1), 24–33.

<https://doi.org/10.14359/14246>

Scrivener, K. L., Juilland, P., & Monteiro, P. J. M. (2015). Advances in understanding hydration of Portland cement. *Cement and Concrete Research*, 78, 38–56. <https://doi.org/10.1016/j.cemconres.2015.05.025>

Šmilauer, V., Edelmannová, J., & Reiterman, P. (2026). Isothermal calorimetry database of 65 cements with analytical approximations. *Ceramics-Silikáty*, 70(1), 25–33. <https://doi.org/10.13168/cs.2025.0048>

Snelson, D. G., Wild, S., & O'Farrell, M. (2008). Heat of hydration of Portland cement–metakaolin–fly ash (PC–MK–PFA) blends. *Cement and Concrete Research*, 38(6), 832–840. <https://doi.org/10.1016/j.cemconres.2008.01.004>

Taylor, H. F. W. (1997). *Cement chemistry* (2nd ed.). Thomas Telford Publishing.

Wang, X.-Y. (2014). Properties prediction of ultra high performance concrete using blended cement hydration model. *Construction and Building Materials*, 65, 151–160. <https://doi.org/10.1016/j.conbuildmat.2014.04.092>

Wang, X.-Y., Lee, H.-S., Park, K.-B., Kim, J.-J., & Golden, J. S. (2010). A multi-phase kinetic model to simulate hydration of slag–cement blends. *Construction and Building Materials*, 24(12), 2357–2365. <https://doi.org/10.1016/j.conbuildmat.2010.05.003>

Wolterbeek, T. K. T., & Hangx, S. J. T. (2023). The thermal properties of set Portland cements – a literature review in the context of CO₂ injection well integrity. *International Journal of Greenhouse Gas Control*, 128, Article 103909. <https://doi.org/10.1016/j.ijggc.2023.103909>

Zákoutský, J., Tydlitát, V., & Černý, R. (2012). Effect of temperature on the early-stage hydration characteristics of Portland cement: A large-volume calorimetric study. *Construction and Building Materials*, 36, 699–706. <https://doi.org/10.1016/j.conbuildmat.2012.06.025>

APPENDIX

Appendix A

X-Ray Diffraction Characterization Records

A.1 General

X-ray diffraction analysis was conducted to identify the crystalline phases present in the four cement samples used in this study. The XRD results were used to support the interpretation of hydration kinetics, heat generation rate, and cumulative heat release, and to prepare VCCTL simulation inputs.

The XRD characterization was performed for the following samples:

Sample 1: CEM II B-M 42.5 N

Sample 2: CEM II A-M 42.5 N

Sample 3: CEM II A-M 42.5 R

Sample 4: CEM I 52.5 N

The XRD patterns, matched phases, scale factors, and calculated semi-quantitative phase compositions are presented in this appendix.

A.2 XRD Measurement Conditions

The XRD analysis was performed using a PANalytical EMPYREAN diffractometer equipped with Cu K α radiation. The same scan settings were used for all four samples to ensure comparability.

Table A.1: XRD measurement conditions

Parameter	Value
Diffractometer system	PANalytical EMPYREAN
Radiation source	Cu K α
K α 1 wavelength	1.54060 Å
K α 2 wavelength	1.54443 Å
Scan axis	Gonio
Start position	6.0056° 2 θ
End position	89.9856° 2 θ
Step size	0.0260° 2 θ

Parameter	Value
Scan step time	7.1400 s
Scan type	Continuous
PSD mode	Scanning
PSD length	3.35° 2 θ
Divergence slit type	Fixed
Divergence slit size	0.7197°
Specimen length	10.00 mm
Measurement temperature	25.00°C
Generator setting	40 mA, 45 kV
Sample spinning	No

A.3 XRD Diffractograms

The XRD diffractograms of the four samples are presented in Figures A.1 to A.4. These figures show the measured intensity-count response over the scanned 2θ range and the matched reference phase positions.

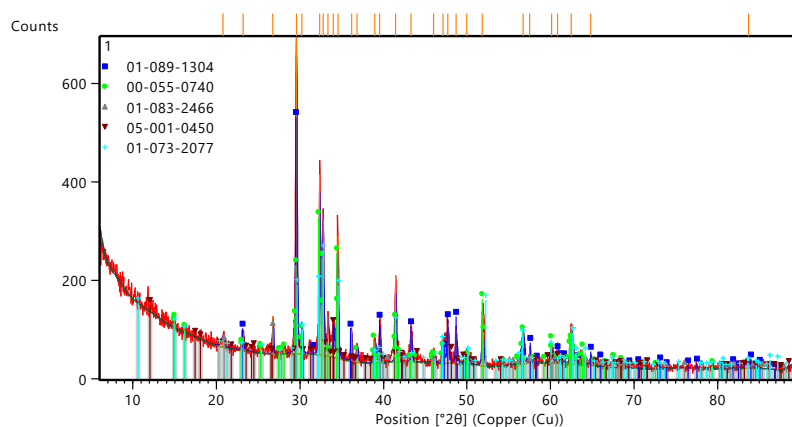


Figure 1: *XRD diffractograms for CEM II B-M 42.5 N*

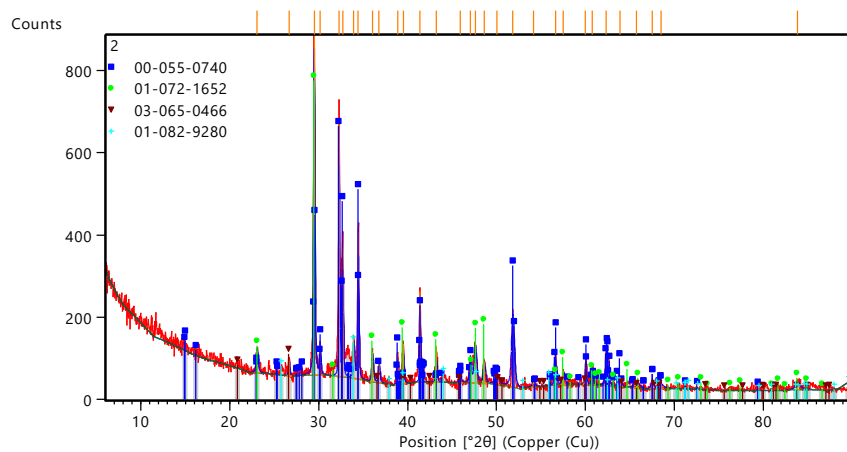


Figure 2: *XRD diffractograms for CEM II A-M 42.5 N*

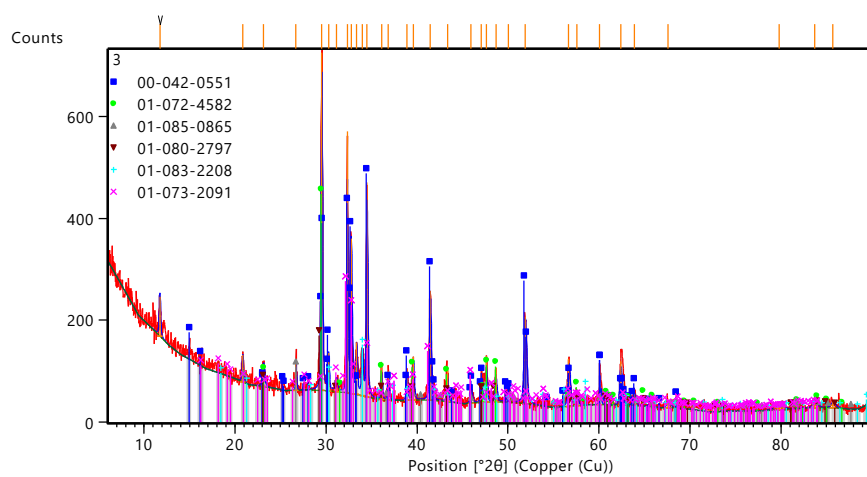


Figure 3: *XRD diffractograms for CEM II A-M 42.5 R*

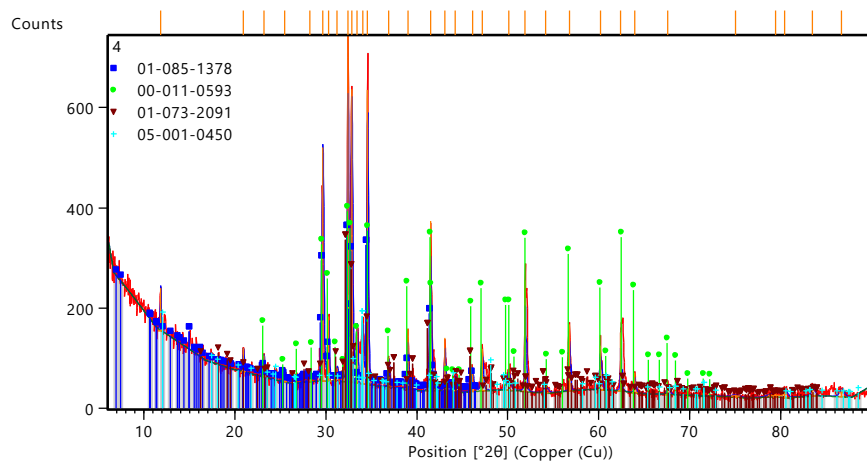


Figure 4: *XRD diffractograms for CEM I 52.5 N*

A.4 XRD Phase Identification Results

The phase identification was performed using PANalytical HighScore Plus software with the ICDD PDF-4+ database. The matched compounds, chemical formulas, scores, and scale factors are listed in Table A.2.

Table A.2: XRD phase identification results for CEM II B-M 42.5 N

Visible	Ref.Code	Score	Compound Name	Displ.[°2θ]	Scale Fac.	Chem. Formula
*	01-085-1378	36	Calcium Oxide Silicate	0.000	0.406	Ca ₃ (SiO ₄)O
*	00-011-0593	36	Calcium Magnesium Aluminum Silicate	0.000	0.458	54CaO ·16SiO ₂ · Al ₂ O ₃ · MgO
*	01-073-2091	10	Calcium Silicate	0.000	0.381	Ca ₂ (SiO ₄)
*	05-001-0450	14	Calcium Aluminum Oxide	0.000	0.182	Ca ₂ Al ₂ O ₅

Table A.3: XRD phase percentage for CEM II B-M 42.5 N

Compound	Scale Factor	Phase % (approx)
(MgO.O ₃ CaO.97CO ₃)	0.691	43.87%
Ca ₃ SiO ₅	0.403	25.59%
SiO ₂	0.079	5.02%
Ca ₂ Al ₂ O ₅	0.094	5.97%
Ca ₃ (SiO ₄)O	0.308	19.55%

Table A.4: XRD phase identification results for CEM II A-M 42.5 N

Visible	Ref.Code	Score	Compound Name	Displ. [°2θ]	Scale Fac.	Chem. Formula
*	00-0420551	61	Calcium Silicate	0.000	0.596	Ca ₃ SiO ₅
*	01-0724582	49	Calcium Carbonate	0.000	0.525	Ca(CO ₃)
*	01-0850865	16	Silicon Oxide	0.000	0.064	SiO ₂
*	01-0802797	17	Calcium Carbonate	0.000	0.146	Ca(CO ₃)
*	01-0832208	15	Calcium Aluminum Silicate	0.000	0.136	Ca ₃ Al ₂ Si ₃ O ₁₂
*	01-0732091	18	Calcium Silicate	0.000	0.298	Ca ₂ (SiO ₄)

Table A.5: XRD phase percentage for CEM II A-M 42.5 N

Compound	Scale Factor	Phase % (Weight Approx)
Ca ₃ SiO ₅	0.685	41.49%
Ca(CO ₃)	0.806	48.82%
SiO ₂	0.061	3.69%
(Fe _{0.65} Mg _{0.35} CO ₃)	0.099	6.00%

Table A.6: XRD phase identification results for CEM II A-M 42.5 R

Visible	Ref.Code	Score	Compound Name	Displ. [°2θ]	Scale Fac.	Chem. Formula
*	00-042-0551	61	Calcium Silicate	0.000	0.596	Ca ₃ SiO ₅
*	01-072-4582	49	Calcium Carbonate	0.000	0.525	Ca(CO ₃)

*	01-085-0865	16	Silicon Oxide	0.000	0.064	SiO ₂
*	01-080-2797	17	Calcium Carbonate	0.000	0.146	Ca(CO ₃)
*	01-083-2208	15	Calcium Aluminum Silicate	0.000	0.136	Ca ₃ Al ₂ Si ₃ O ₁₂
*	01-073-2091	18	Calcium Silicate	0.000	0.298	Ca ₂ (SiO ₄)

Table A.7: XRD phase percentage for CEM II A-M 42.5 R

Compound	Scale Factor	Phase % (Approx)
Ca ₃ SiO ₅	0.596	33.77%
Belite (Ca ₂ SiO ₄)	0.298	16.88%
Ca(CO ₃)	0.671*	38.02%
Ca ₃ Al ₂ Si ₃ O ₁₂	0.136	7.71%
SiO ₂	0.064	3.62%

Table A.8: XRD phase identification results for CEM I 52.5 N

Visible	Ref.Code	Score	Compound Name	Displ. [°2]	Scale Fac.	Chem. Formula
*	01-085-1378	36	Calcium Oxide Silicate	0.000	0.406	Ca ₃ (SiO ₄)O
*	00-011-0593	36	Magnesium Aluminum Silicate	0.000	0.458	54CaO ·16SiO ₂ · Al ₂ O ₃ · MgO
*	01-073-2091	10	Calcium Silicate	0.000	0.381	Ca ₂ (SiO ₄)
*	05-001-0450	14	Calcium Aluminum Oxide	0.000	0.182	Ca ₂ Al ₂ O ₅

Table A.9: XRD phase percentage for CEM I 52.5 N

Compound	Scale Factor	Phase % (Approx)
$\text{Ca}_3(\text{SiO}_4)\text{O}$	0.406	28.45%
$54\text{CaO} \cdot 16\text{SiO}_2 \cdot \text{Al}_2\text{O}_3 \cdot \text{MgO}$	0.458	32.10%
$\text{Ca}_2(\text{SiO}_4)$	0.381	26.70%
$\text{Ca}_2\text{Al}_2\text{O}_5$	0.182	12.75%

Calculation of XRD phase percentage for CEM II A-M 42.5 R

Compound	Scale Factor
Ca_3SiO_5	0.596
CaCO_3	0.671
SiO_2	0.064
$\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	0.136
$\text{Ca}_2(\text{SiO}_4)$	0.298

Total scale factor = $0.596 + 0.671 + 0.064 + 0.136 + 0.298 = 1.765$

$\text{CaCO}_3 = (0.671 \div 1.765) \times 100 = 38.02\%$

Similarly,

$\text{Ca}_3\text{SiO}_5 = 33.77\%$

$\text{CaCO}_3 = 38.02\%$

$\text{SiO}_2 = 3.62\%$

$\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} = 7.71\%$

$\text{Ca}_2(\text{SiO}_4) = 16.88\%$

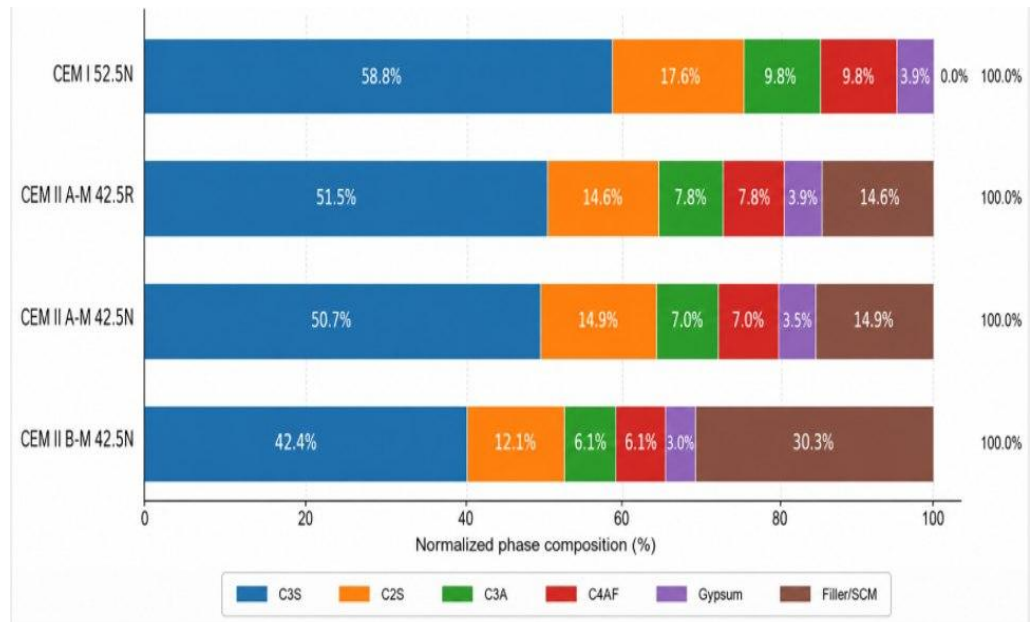


Figure A.7: Semi-quantitative XRD phase composition of the four samples

Appendix B

Calibration Constant and Heat-Loss Correction

B.1 General

The semi-adiabatic calorimeter was calibrated before conducting the hydration heat tests. Calibration was necessary to determine the system's effective heat capacity and heat-loss coefficient. These parameters were used to convert the measured temperature rise into the heat generation rate and the cumulative heat release.

B.2 Effective Heat Capacity of the Calorimeter System

The effective heat capacity was determined by electrical heating calibration. A heater of known power was used to heat water inside the calorimeter system for a known duration. The effective heat capacity was calculated using:

$$C = \frac{Pt}{\Delta T}$$

Where,

C=effective heat capacity, J/K

P=heater power, W

t=heating duration, s

ΔT =temperature rise, K

Table B.1: Calibration data for effective heat capacity

Parameter	Value	Unit
Water mass	404	g
Specific heat of water	4.184	J/g-K
Heater power	500	W
Initial temperature	19.7	°C
Final temperature	60.0	°C
Temperature rise	40.3	K
Heating duration	248	s
Effective heat capacity	3076.9	J/K
Insulator heat capacity	1386.6	J/K

The heat input was:

$$C = \frac{500 \times 248}{40.3} = 3076.9 \text{ J/K}$$

The heat capacity of water was:

$$C_w = m_s = 404 \times 4.184 = 1690.3 \text{ J/K}$$

The insulator heat capacity was:

$$C_{\text{insulator}} = 3076.9 - 1690.3 = 1386.6 \text{ J/K}$$

B.3 Heat-Loss Coefficient from Cooling Experiment

The heat-loss coefficient was determined using a cooling experiment. The vial was heated above ambient temperature and then allowed to cool naturally. The temperature decay was fitted using the lumped thermal model.

The governing equation is:

$$\frac{dT}{dt} = -\frac{K_l}{C}(T - T_{\text{amb}})$$

By integrating, $(T(t) - T_{amb}) = (T_0 - T_{amb})e^{-\frac{k_L}{C}t}$.

Fit an exponential to the measured cooling curve to get $\alpha = \frac{K_L}{C}$.

Then, $K_L = \alpha C$

Table B.2: Cooling experiment data

Parameter	Value	Unit
Initial temperature, (T_0)	59.9	°C
Final temperature, ($T(t)$)	58.4	°C
Ambient temperature, (T_{amb})	23.1	°C
Cooling duration, (t)	3600	s
Effective heat capacity, (C)	3076.9	J/K

$$K_L = -\frac{3076.9}{3600} \ln \frac{58.4 - 23.1}{59.9 - 23.1}$$

$$= 0.0355 \text{ W/K}$$

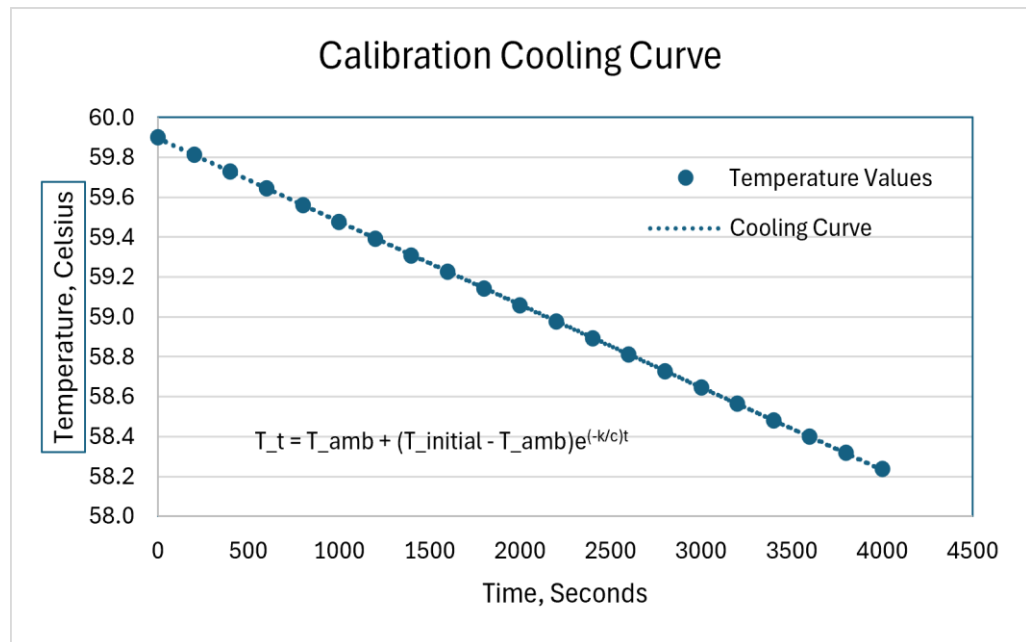


Figure B.2: Cooling curve used for heat-loss coefficient calculation

Appendix B.4 Apparatus Used

Table B.3: Apparatus used for calibration and hydration heat measurement

Sl. No.	Apparatus	Purpose
1	Insulated calorimeter container	To reduce heat loss
2	500 W, 220 V heater	Electrical calibration
3	Temperature sensor	Measurement of sample temperature
4	Temperature-humidity meter	Ambient condition monitoring
5	Cement paste vial	Holding sealed cement paste
6	Data logger	Continuous temperature recording
7	Computer	Data storage and MATLAB processing
8	Weighing balance	Measuring cement and water

Figure B.3: Apparatus used for semi-adiabatic calorimetry and calibration.



Appendix C

Heat Generation and Cumulative Heat Calculation

C.1 Heat Balance Equation

The reaction heat flow was calculated from the measured temperature rise and heat-loss correction. The lumped heat balance equation is:

$$C \frac{dT}{dt} = \dot{q}_{reaction}(t) - K_l(T(t) - T_{amb}(t))$$

Where,

$C = mc = \text{effective capacity of sample (J/k)}$

$K_l = \text{heat - loss coefficient (W/K)}$

$\dot{q}_{reaction} = \text{instantaneous heat flow from reaction (W)}$

Rearrange the above equation to get instantaneous heat flow,

$$\dot{q}_{reaction}(t) = C \frac{dT}{dt} + K_l(T(t) - T_{amb}(t))$$

By integrating, we will found total heat from time elapsing 0 to t1:

$$Q = \int_0^{t_1} \dot{q}_{reaction}(t)dt = C \int_0^{t_1} dT + \int_0^{t_1} K_l(T(t) - T_{amb}(t))dt$$

where:

$\dot{q}_{reaction}(t) = \text{instantaneous heat generation rate, W}$

$C = \text{effective heat capacity of the system, } \frac{J}{K}$

$\frac{dT}{dt} = \text{rate of temperature change, K/s}$

$K_l = \text{heat - loss coefficient, W/K}$

$T(t) = \text{sample temperature, } ^\circ\text{C}$

$T_{amb}(t) = \text{ambient temperature, } ^\circ\text{C}$

C.2 Heat Generation Rate Per Unit Mass

The heat generation rate per unit mass of cement was calculated as:

$$q(t) = \frac{\dot{q}_{reaction}(t)}{m}$$

where:

$q(t)$ = heat generation rate per unit mass of cement

m = mass of cement used in the paste specimen

The final heat generation rate was expressed in J – min/g.

C.3 Cumulative Heat Release

The cumulative heat release was calculated by integrating the heat generation rate curve:

$$Q(t) = \int_0^t q(t)dt$$

Simpson's method of numerical integration has been used to quantify heat release:

Simpson's 1/3 Rule

Divide the time interval into **n even subintervals** (n must be even):

Step size:

$$h = \frac{t_n - t_0}{n} ; n = 2$$

Discrete values:

$$q_0, q_1, q_2, \dots, q_n$$

Formula:

$$Q \approx \frac{h}{3} [q_0 + 4(q_1 + q_3 + \dots + q_{n-1}) + 2(q_2 + q_4 + \dots + q_{n-2}) + q_n]$$

C.4 Degree of Hydration

The degree of hydration is defined by $r(t)$ of cementitious material defined as the fraction of cement that has already hydrated:

$$r(t) = \frac{\text{amount of cement that has reacted at time } t}{\text{total amount of time at } t = 0}$$

A practical method to estimate the degree of hydration is based on the exothermic character of the hydration process:

$$r(t) = \frac{Q(t)}{Q_{max}}$$

Where $Q(t)$ is the total heat developed at time t (in J/g), and $Q_{\max}$ is the total heat development corresponding to complete hydration (in J/g).

The total heat $Q(t)$ can be calculated as

$$Q(t) = \int_0^t q(t) dt$$

Where $q(t)$ is the heat production (in J/g per h)

In ref. 2, a new hydration model is developed, yielding a heat production q depending on the actual degree of hydration $r(t)$ and the actual temperature $\theta(t)$

$$q(t) = q[r(t), \theta(t)]$$

From the third consecutive equation above, we can establish the following relationships:

$$\frac{dQ(t)}{dt} = q(t) = q[r(t), \theta(t)]$$

Now, differentiating the second consecutive equation and the introduction of equation yield,

$$dr(t) = \frac{dQ(t)}{Q_{\max}} = \frac{q[r(t), \theta(t)]}{Q_{\max}} dt$$

Hence, the value of r can be updated by substituting the above equation,

$$r(t + dt) = r(t) + dr(t) \text{ or}$$

$$r(t + \Delta t) = r(t) + \Delta r$$

Appendix D

MATLAB Gaussian and Polynomial Fitting Results

D.1 General

The processed heat generation rate and cumulative heat release data were analyzed using MATLAB. A two-component Gaussian model was used to represent the heat generation rate curve, while a sixth-order polynomial model was used to fit the cumulative heat release curve.

The two-component Gaussian model is expressed as:

$$q(t) = A_1 e^{\left\{-\left(\frac{t-\mu^1}{\sigma^1}\right)^2\right\}} + A_2 e^{\left\{-\left(\frac{t-\mu^2}{\sigma^2}\right)^2\right\}}$$

where:

A_1 = amplitude of the first peak

μ_1 = time of the first peak

σ_1 = width parameter of the first peak

A_2 = amplitude of the second peak

μ_2 = time of the second peak

σ_2 = width parameter of the second peak

D.2 CEM II B-M 42.5 N

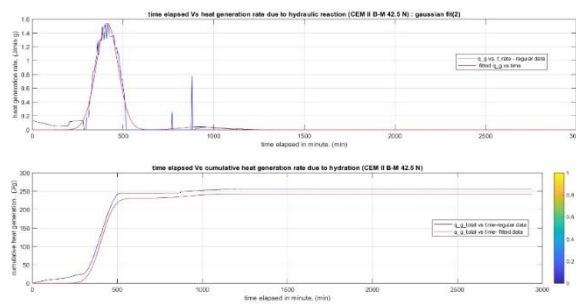


Figure D.1: Heat generation rate & Cumulative Heat: CEM II B-M 42.5 N
(Gaussian Fit 2)

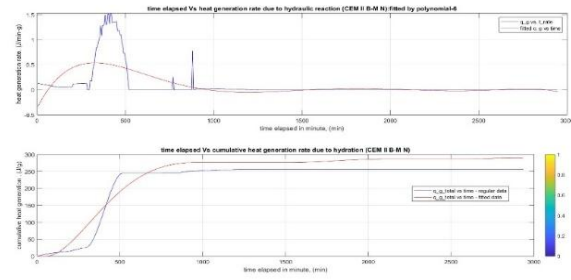


Figure D.2: Heat generation rate & Cumulative Heat: CEM II B-M 42.5 N
(Polynomial 06)

D.3 CEM II A-M 42.5 N

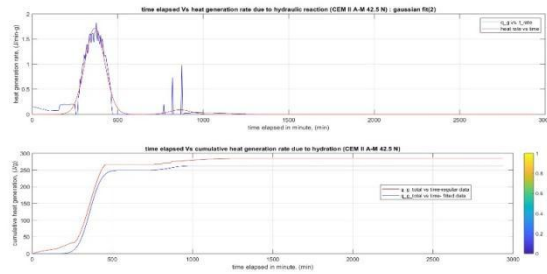


Figure D.1: Heat generation rate & Cumulative Heat: CEM II A-M 42.5 N
(Gaussian Fit 2)

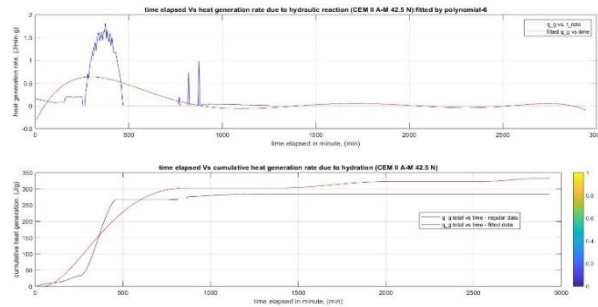


Figure D.2: Heat generation rate & Cumulative Heat: CEM II A-M 42.5 N
(Polynomial 06)

D.4 CEM II A-M 42.5 R

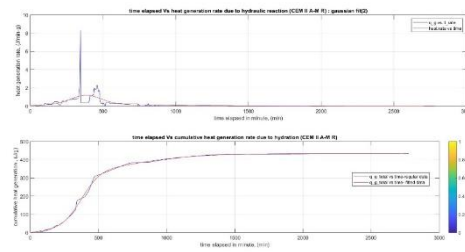


Figure D.1: Heat generation rate & Cumulative Heat: CEM II A-M 42.5 R
(Gaussian Fit 2)

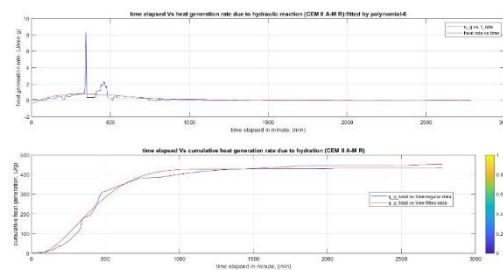


Figure D.2: Heat generation rate & Cumulative Heat: CEM II A-M 42.5 R
(Polynomial 06)

D.5 CEM I 52.5 N

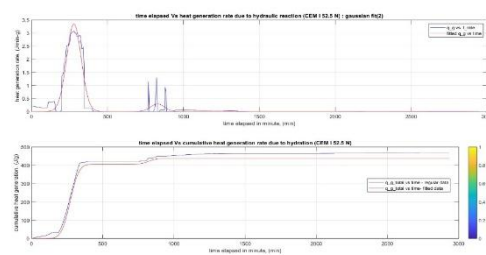


Figure D.1: Heat generation rate & Cumulative Heat: CEM I 52.5 N
(Gaussian Fit 2)

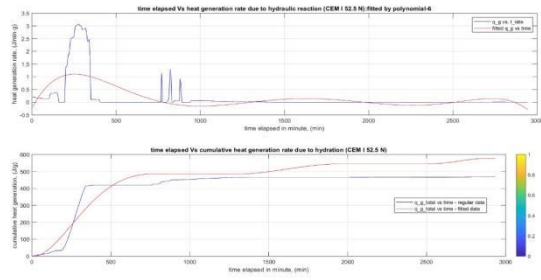


Figure D.2: Heat generation rate & Cumulative Heat: CEM I 52.5 N
(Polynomial 06)

D.6 Summary of Gaussian Model Parameters

Table D.1: Gaussian model parameters and cumulative heat release

Parameter	CEM I 52.5N	CEM II A-M 42.5N	CEM II B-M 42.5N	CEM II A-M 42.5R	Unit
G ₁ Amplitude (A ₁)	~0.40–0.45	~0.15–0.20	~0.12–0.16	~0.30–0.40	J/min-g
G ₁ Peak Time (μ ₁)	~220–260	~280–320	~320–360	~230–270	min
G ₂ Amplitude (A ₂)	~2.8–3.3	~1.5–2.0	~1.3–1.6	~2.2–2.6	J/min-g
G ₂ Peak Time (μ ₂)	~280–340	~400–460	~450–510	~310–370	min
Cumuli. heat (terminal)	~440–480	~260–285	~245–265	~430–460	J/g
Fit order	Gaussian 2	Gaussian 2	Gaussian 2	Gaussian 2	

Appendix E

VCCTL Simulation Results

E.1 General

The NIST Virtual Cement and Concrete Testing Laboratory was used to simulate the degree of hydration and heat release of the four cement types. The VCCTL simulations were carried out under isothermal conditions using XRD-derived clinker phase information, water-to-cement ratio, and particle characteristics as input parameters.

E.2 CEM II B-M 42.5 N

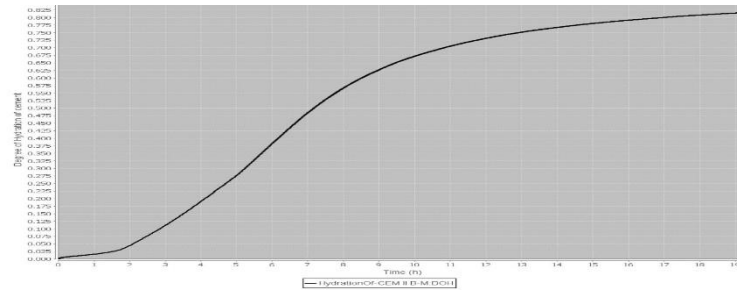


Figure E.2: Degree of Hydration of Cement by VCCTL Software: CEM II B-M 42.5 N

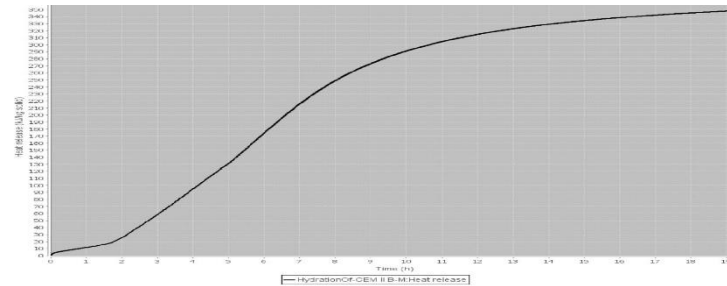


Figure E.2: Heat Generation rate of Cement by VCCTL Software: CEM II B-M 42.5 N

E.3 CEM II A-M 42.5 N

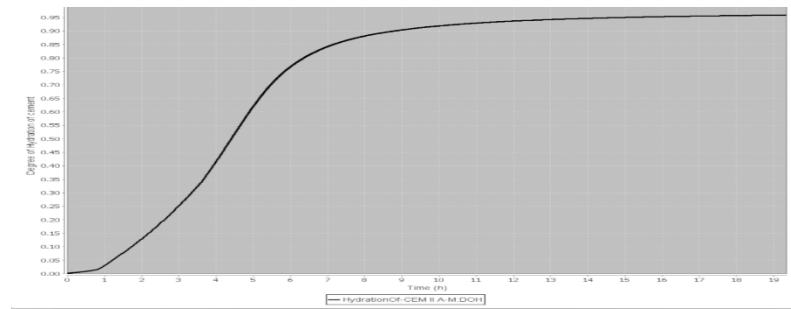


Figure E.2: Degree of Hydration of Cement by VCCTL Software: CEM II A-M 42.5 N

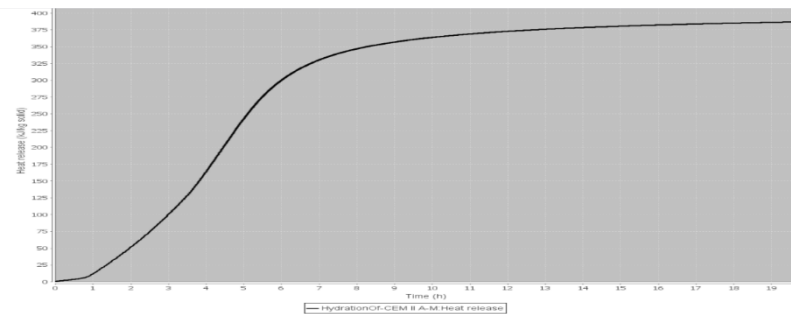


Figure E.2: Heat Generation rate of Cement by VCCTL Software: CEM II A-M 42.5 N

E.4 CEM II A-M 42.5 R

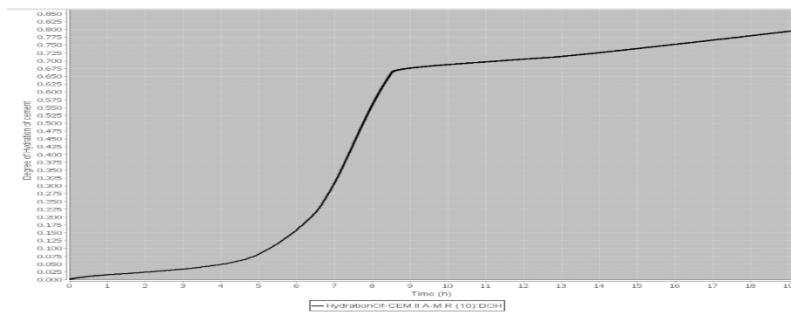


Figure E.2: Degree of Hydration of Cement by VCCTL Software: CEM II A-M 42.5 R

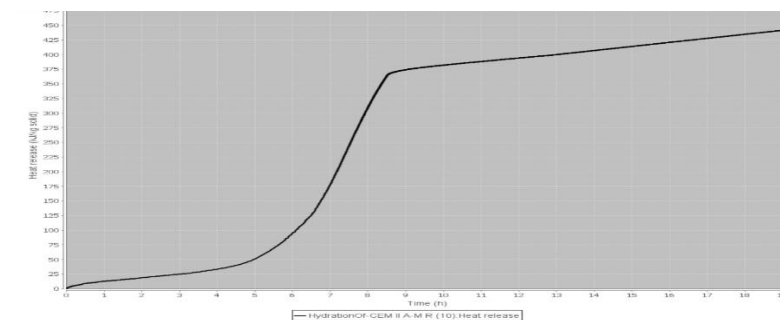


Figure E.2: Heat Generation rate of Cement by VCCTL Software: CEM II A-M 42.5 R

E.5 CEM I 52.5 N

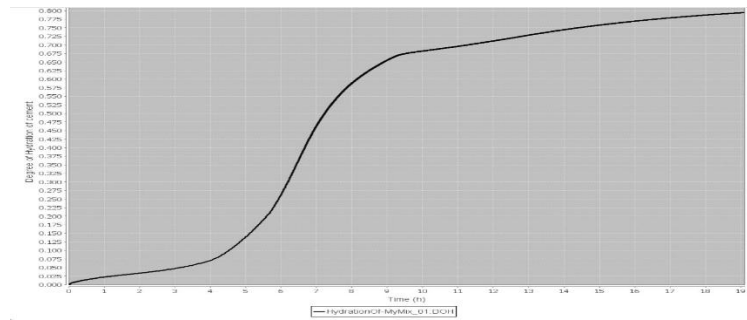


Figure E.2: Degree of Hydration of Cement by VCCTL Software: CEM I 52.5 N

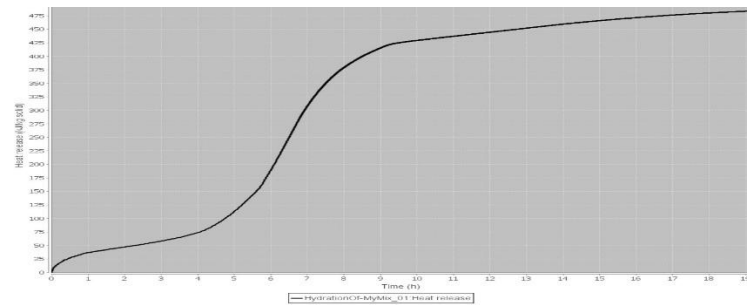


Figure E.2: Heat Generation rate of Cement by VCCTL Software: CEM I 52.5 N

E.6 VCCTL Simulation Summary

Table E.1: VCCTL simulation summary at 19 hours

Cement/mix	VCCTL DOH at 19 h	VCCTL heat release at 19 h	Condition
CEM II B-M 42.5 N	0.825	348 kJ/kg	Isothermal
CEM II A-M 42.5 N	0.950	390 kJ/kg	Isothermal
CEM II A-M 42.5 R	0.800	445 kJ/kg	Isothermal
CEM I 52.5 N	0.81	480 kJ/kg	Isothermal

Appendix K

MATLAB Algorithm for Semi-Adiabatic Hydration Heat Analysis

K.1 General

This appendix presents the MATLAB algorithm used for processing semi-adiabatic calorimetry data. The algorithm converts raw temperature-time data into heat generation rate and cumulative heat release. It also performs Gaussian fitting, sixth-order polynomial fitting, peak heat analysis, and graphical output generation.

The algorithm was applied to all four cement samples tested in this study, namely CEM I 52.5 N, CEM II A-M 42.5 N, CEM II A-M 42.5 R, and CEM II B-M 42.5 N.

K.2 Algorithm Description

Input data:

1. Raw Excel file containing ambient temperature, cement paste temperature, and elapsed time.
2. VCCTL output file containing hydration time and cumulative heat release.
3. Cement mass and water mass used in the paste.
4. Calibrated heat capacity and heat-loss coefficient.

Output data:

1. Interpolated temperature-time data.
2. Heat generation rate, $q(t)$.
3. Cumulative heat release, $Q(t)$.
4. Gaussian-fitted heat generation curve.
5. Polynomial-fitted heat generation curve.
6. Peak heat generation rate and corresponding time.
7. Graphs for heat generation rate and cumulative heat release.

K.3 Step-by-Step Algorithm

Algorithm K.1: Semi-Adiabatic Hydration Heat Processing Algorithm

1. Start the program.
2. Read raw temperature data from Excel.
3. Extract:
 - Ambient temperature, (T_{amb})
 - Cement pastes temperature (T_s)
 - Elapsed time, (t)
4. Replace zero or missing time values with interpolated values.
5. Fill missing temperature values using linear interpolation.
6. Sort the data according to elapsed time.
7. Remove duplicate time values.
8. Generate a uniform time vector at 5-minute intervals.
9. Interpolate ambient temperature and sample temperature onto the uniform time vector.
10. Read VCCTL heat release data.
11. Extract VCCTL time and heat release values.
12. Convert VCCTL time from hours to minutes.
13. Interpolate VCCTL heat release data to match the experimental time vector.
14. Calculate the degree of hydration
15. Calculate variable heat capacity of the cement paste system
16. Calculate the rate of temperature rise
17. Calculate the temperature difference between sample and ambient
18. Calculate total heat generation rate
19. Convert total heat generation rate to heat generation rate per gram of cement
20. Remove negative heat rate values by setting them equal to zero.
21. Fit the heat generation rate curve using a two-component Gaussian model
22. Fit the heat generation rate curve using a sixth-order polynomial:
$$Q(t) = c_0 + c_1t + c_2t^2 + c_3t^3 + c_4t^4 + c_5t^5 + c_6t^6$$
23. Calculate cumulative heat release by numerical integration:

$$Q(t) = \int_0^t q(t)dt$$

24. Calculate cumulative heat for both raw data and fitted data.
25. Determine the maximum heat generation rate.
26. Determine the time corresponding to the maximum heat generation rate.
27. Plot:
 - Heat generation rate versus time
 - Gaussian fitted curve
 - Polynomial fitted curve
 - Cumulative heat release from raw data
 - Cumulative heat release from fitted data
28. Export processed results and figures.
29. End the program.

Algorithm for Semi-Adiabatic Hydration Heat Analysis



K.4 MATLAB Code Used for Hydration Heat Analysis

This appendix presents the complete MATLAB source code developed for the hydration analysis of CEM II A-M 42.5 R cement under semi-adiabatic conditions. The scripts are listed in functional call order — from the top-level analysis driver down to individual utility functions — to facilitate readability and reproducibility.

Function call hierarchy:

recordeddata3.m — Raw Data Reader

Reads ambient and sample temperature columns together with elapsed time from the raw calorimetry Excel file. Replaces zero and missing entries using linear interpolation.

```
function [known_temp, final_time] = recordeddata3()
% =====
% recordeddata3.m
% Reads raw calorimetry data from Excel and removes
% missing/zero entries using linear interpolation.
% =====

[num] = readtable("CEM II A-M 42.5 R.xlsx");

known_temp = num(:, [1 2]); % columns: ambient temp, sample temp

final_time = num(:, 9);
final_time.Var9(final_time.Var9 == 0) = NaN;

%% Clear missing data
known_temp = fillmissing(known_temp, 'linear');
final_time = fillmissing(final_time, 'linear');

end
```

Data_analysing3.m — Temperature Interpolation

Calls recordeddata3() and resamples both temperature signals onto a uniform 5-minute grid using linear interpolation with extrapolation at boundaries.

```
function [tamb_new, ts_new, t_new] = Data_analysing3()
% =====
% Data_analysing3.m
% Interpolates ambient and sample temperatures onto a
% uniform 5-minute time grid using linear interpolation.
```

```

% =====

[known_temp, final_time] = recordeddata3();
known_temp = table2array(known_temp);
final_time = table2array(final_time);

tamb_known = known_temp(:, 1);
ts_known = known_temp(:, 2);

%% Sort by time
[final_time, idx] = sort(final_time);
tamb_known = tamb_known(idx);
ts_known = ts_known(idx);

%% Interpolation onto uniform grid (every 5 minutes)
t_new = 0:5:final_time(end);

tamb_new = interp1(final_time, tamb_known, t_new, 'linear', 'extrap');
ts_new = interp1(final_time, ts_known, t_new, 'linear', 'extrap');

t_new = t_new.';
tamb_new = tamb_new.';
ts_new = ts_new.';

end

```

C_adjustment3.m — Heat Capacity Correction

Reads the reference heat-release profile from the VCCTL-compatible Excel file, matches it to the calorimetric time grid, and computes the degree-of-hydration ratio r . This ratio is used in `heat_gen_rate3()` to dynamically adjust the apparent heat capacity C_a according to the De Schutter–Taerwe model.

```

function [heat_release_2, r] = C_adjustment3()
% =====
% C_adjustment3.m
% Reads calorimeter heat-release data, interpolates onto
% the calorimetric time grid, and computes the degree-of-
% hydration ratio r (used to adjust heat capacity C_a).
% =====

[data] = readtable("HydrationOf-MyMix_Sample 03 - CEM II A-M 42.5 R.xlsx");
time = data(:, 2);

```

```

heat_release = data(:, 5);

time      = table2array(time);
heat_release = table2array(heat_release);

%% Restrict to first 19 hours, convert to minutes
time = time(time <= 19);
time = time * 60;
[time, idx] = sort(time);
heat_release = heat_release(idx);

%% Match onto calorimetric time vector
[tamb_new, ts_new, t_new] = Data_analysing3();
t_new_c = t_new(t_new < 1140);

heat_release_2 = interp1(time, heat_release, t_new_c, 'linear');
heat_release_2(229:559) = heat_release_2(end); % extend plateau

%% Compute degree-of-hydration ratio r
r = (1:length(t_new));

for i = 1:length(t_new)
    r(i) = heat_release_2(i) / heat_release_2(end);
end
end

```

heat_gen_rate3.m — Heat Generation Rate

Core thermophysical calculation. Applies the semi-adiabatic energy balance — incorporating the degree-of-hydration-dependent heat capacity C_a and the calibration heat-loss constant K_l — to derive the specific heat generation rate q_g (J/min/g cement) at each time step.

```

function [t_new, tamb_new, ts_new, q_g, t_rate, C] = heat_gen_rate3()
% =====
% heat_gen_rate3.m
% Computes the specific heat generation rate q_g (J/min/g)
% from semi-adiabatic calorimetry data using the corrected
% heat-capacity approach (De Schutter & Taerwe model).
% =====

%% Call data analysis function

```

```

[tamb_new, ts_new, t_new] = Data_analysing3();

%% Specimen and vessel parameters
m_c = 139; % cement mass (g)
m_w = 57; % water mass (g)
m_total = m_c + m_w;

cp_c = 0.84; % specific heat of cement (J/g/K)
cp_w = 4.18; % specific heat of water (J/g/K)

cp_paste = (cp_c*m_c + cp_w*m_w) / m_total;
Cp = cp_paste * m_total; % heat capacity of paste (J/K)
Cv = 1386.6; % heat capacity of vessel (J/K)
C = Cp + Cv; % total system heat capacity (J/K)

%% Degree-of-hydration correction for C_a
[heat_release_2, r] = C_adjustment3();
C_a = 1160.67 .* (1.5 - 0.5 .* r);

%% Calibration constant
Kl = 2.13; % heat-loss constant (J/min/K), from calibration

%% Temperature derivatives and heat-rate calculation
dT_dt = diff(ts_new) ./ diff(t_new); % dT/dt (K/min)
t_rate = t_new(1:end-1);
delta_t = ts_new(1:end-1) - tamb_new(1:end-1); % temperature excess (K)

%% Total heat rate and specific heat generation rate
q = C_a(1:end-1) .* dT_dt + Kl .* delta_t; % J/min
q_g = q / m_c; % J/min/g cement

end

```

createFit_Gauss3.m — Two-Component Gaussian Fitting

Fits a gauss2 model to q_g data. The first Gaussian (G1) represents the early aluminate (C₃A) reaction peak; the second (G2) captures the dominant silicate (C₃S/C₂S) hydration peak. Cumulative heat is computed by Simpson's rule for both the measured and fitted curves, and the results are displayed as a two-panel figure.

```

function [q_g, q_g_fit, t_fit, fitresult, gof] = createFit_Gauss3()
% =====

```

```

% createFit_Gauss3.m
% Fits a two-component Gaussian model (gauss2) to the
% heat generation rate data of CEM II A-M 42.5 R and
% computes cumulative heat by Simpson's rule.
% G1 captures the early aluminate peak; G2 captures the
% dominant silicate (C3S) hydration peak.
% =====

[t_new, tamb_new, ts_new, q_g, t_rate] = heat_gen_rate3();

%% Pre-process: clip negatives and trim leading point
q_g(q_g < 0) = 0;
t_rate = t_rate(2:end);
q_g = q_g(2:end);

%% Gaussian fit (gauss2)
[xData, yData] = prepareCurveData(t_rate, q_g);

ft = fitype('gauss2');
opts = fitoptions('Method', 'NonlinearLeastSquares');
opts.Display = 'Off';
opts.Lower = [-Inf -Inf 0 -Inf -Inf 0];
opts.StartPoint = [9.62645083932855 345 121.411402695017 ...
    1.0616919313023 680 206.268279960366];

[fitresult, gof] = fit(xData, yData, ft, opts);
q_g_fit = fitresult(xData);
t_fit = xData;

%% Cumulative heat — raw data (Simpson's rule)
q_g_total = zeros(1, length(t_fit)-2);
for i = 1:(length(t_fit)-2)
    increment = 5*(q_g(i) + 4*q_g(i+1) + q_g(i+2)) / 6;
    q_g_total(i) = increment + (i > 1) * q_g_total(i-1);
end

%% Cumulative heat — fitted data (Simpson's rule)
q_g_total2 = zeros(1, length(t_fit)-2);
for i = 1:(length(t_fit)-2)
    increment2 = 5*(q_g_fit(i) + 4*q_g_fit(i+1) + q_g_fit(i+2)) / 6;
    q_g_total2(i) = increment2 + (i > 1) * q_g_total2(i-1);
end

```

```

%% Plot: heat generation rate
figure('Name', 'CEM II A-M R – Gaussian Fit');
subplot(2,1,1);
plot(fitresult, xData, yData, 'b-');
legend('Measured q_g', 'Gauss2 fit', 'Location', 'NorthEast', 'Interpreter', 'none');
title('Heat Generation Rate – CEM II A-M 42.5 R (Gaussian fit, 2 components)');
xlabel('Time elapsed (min)');
ylabel('Heat generation rate (J/min/g)');
grid on;

%% Plot: cumulative heat release
subplot(2,1,2);
plot(t_fit(1:end-2), q_g_total, 'b-', ...
      t_fit(1:end-2), q_g_total2, 'r-');
legend('Measured (cumulative)', 'Fitted (cumulative)', ...
      'Location', 'NorthEast', 'Interpreter', 'none');
title('Cumulative Heat Release – CEM II A-M 42.5 R');
xlabel('Time elapsed (min)');
ylabel('Cumulative heat generation (J/g)');
grid on;

end

```

createFit_Polynomial3.m — 6th-Order Polynomial Fitting

Fits a normalised 6th-order polynomial (poly6) to q_g data to provide a smooth, analytically integrable expression for use in numerical thermal simulations. Cumulative heat is integrated by Simpson's rule, and both rate and cumulative curves are displayed in a two-panel figure.

```

function [q_g_total, q_g, q_g_fit, t_fit, fitresult, gof] = createFit_Polynomial3()
% =====
% createFit_Polynomial3.m
% Fits a 6th-order polynomial to the heat generation rate
% data of CEM II A-M 42.5 R and computes cumulative heat
% by Simpson's rule for both raw and fitted curves.
% =====

[t_new, tamb_new, ts_new, q_g, t_rate] = heat_gen_rate3();

%% Pre-process: clip negatives and trim leading point
q_g(q_g < 0) = 0;

```

```

t_rate = t_rate(2:end);
q_g    = q_g(2:end);

%% Polynomial-6 fit
[xData, yData] = prepareCurveData(t_rate, q_g);

ft = fitype('poly6');
[fitresult, gof] = fit(xData, yData, ft, 'Normalize', 'on');

q_g_fit = fitresult(xData);
t_fit   = xData;
q_g_fit(q_g_fit < 0) = 0; % physical constraint

%% Cumulative heat — raw data (Simpson's rule)
q_g_total = zeros(1, length(t_fit)-2);
for i = 1:(length(t_fit)-2)
    increment = 5*(q_g(i) + 4*q_g(i+1) + q_g(i+2)) / 6;
    q_g_total(i) = increment + (i > 1) * q_g_total(i-1);
end

%% Cumulative heat — fitted data (Simpson's rule)
q_g_total2 = zeros(1, length(t_fit)-2);
for i = 1:(length(t_fit)-2)
    increment2 = 5*(q_g_fit(i) + 4*q_g_fit(i+1) + q_g_fit(i+2)) / 6;
    q_g_total2(i) = increment2 + (i > 1) * q_g_total2(i-1);
end

%% Plot: heat generation rate
figure('Name', 'CEM II A-M R – Polynomial Fit');
subplot(2,1,1);
plot(fitresult, xData, yData, 'b-');
legend('Measured q_g', 'Poly-6 fit', 'Location', 'NorthEast', 'Interpreter', 'none');
title(['Heat Generation Rate – CEM II A-M 42.5 R ' ...
    '(6th-order polynomial fit)']);
xlabel('Time elapsed (min)');
ylabel('Heat generation rate (J/min/g)');
grid on;

%% Plot: cumulative heat release
subplot(2,1,2);
plot(t_fit(1:end-2), q_g_total, 'b-', ...
    t_fit(1:end-2), q_g_total2, 'r-');

```



```
legend('Measured (cumulative)', 'Fitted (cumulative)', ...  
      'Location', 'NorthEast', 'Interpreter', 'none');  
title('Cumulative Heat Release – CEM II A-M 42.5 R');  
xlabel('Time elapsed (min)');  
ylabel('Cumulative heat generation (J/g)');  
grid on;  
  
end
```